Multi-Omics Integration Analysis - MOFA

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Contents

Practical overview **1**

1. Package installation and loading **2**

2. Load the RNA-seq, methylation, and metadata files **3**

3. Metadata filtering and sample alignment **4**

4. Normalize and filter the RNA-seq data **5**

5. Transform and filter the methylation data **6**

6. Prepare the multi-omics data views **7**

7. Create, prepare, and train the MOFA model **8**
Construction and training of the MOFA model 8

8. Inspect variance explained by the trained model **22**

9. Plot the MOFA factors by group **25**

10. Plot Factor 1 against Factor 2 - Factor 1 against Factor 3 **26**

11. Inspect RNA-seq feature weights for Factor 1 **28**

12. Relate the top RNA-seq feature to Factor 1-2-3 (Scatter plots) **32**

13. Visualize the top features driving Factor 1 **39**

Practical overview

This R Markdown document preserves the workflow and execution order of the supplied MOFA.R script, applied in the same dataset we analysed previously. This document contains detailed descriptions of the code cells and the resulting outputs in relation to the dataset used in this practical. It provides an overview of the multi-omics integration process performed using `Multi-Omics Factor Analysis (MOFA)`. The official MOFA2 tutorials, available at `https://biofam.github.io/MOFA2/tutorials.html`, which provides additional examples of model training and downstream analysis.

MOFA utilizes Matrix Factorization framework for the unsupervised integration of multiple omics datasets. It learns an interpretable, low-dimensional representation of the integrated data through a set of latent factors. These factors capture the major sources of variation across the analysed omics modalities and can

subsequently be examined in relation to sample groups, biological characteristics, and influential molecular features.

---

1. Package installation and loading

The required packages are obtained from both CRAN and Bioconductor repositories, with `MOFA2` installed through `BiocManager`. `MOFA2` is the main package used for multi-omics integration in this practical. `basilisk` is used to manage the Python environment and dependencies required by the MOFA2 implementation. In this workflow, this environment is accessed explicitly when the model is trained using `run_mofa(..., use_basilisk = TRUE)`. The stable Bioconductor installation of MOFA2 uses `basilisk` to configure the required Python environment automatically. `edgeR` is used to process the RNA-seq count matrix before integration. `data.table`, `dplyr`, and `tidyr` are loaded for dataframes manipulation.

```
########## MOFA ######################################################################

## Load all the required libraries
# BiocManager::install("MOFA2")


library(MOFA2)
```

```
##
## Attaching package: 'MOFA2'
```

```
## The following object is masked from 'package:stats':
##
##     predict
```

```
library(data.table)
```

```
## data.table 1.18.2.1 using 14 threads (see ?getDTthreads).
```

```
## Latest news: r-datatable.com
```

```
##
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:reshape2':
##
##     dcast, melt
```

```
## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, isoyear, mday, minute, month, quarter, second, wday, week, yday, year
```

```
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
```

```
## The following object is masked from 'package:purrr':
##
##     transpose
```

```
library(ggplot2)
library(dplyr)
library(psych)
```

```
##
## Attaching package: 'psych'

## The following object is masked from 'package:mixOmics':
##
##     pca

## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
library(GGally)
library(ggplot2)
library(ggpubr)
library(tidyr)
library(basilisk)
```

```
## Loading required package: reticulate
```

```
library(edgeR)
```

---

2. Load the RNA-seq, methylation, and metadata files

Boar taint is one of the most extensively studied phenotypes in swine production. It is characterized by an unpleasant urine- or feces-like odor and flavour that is released during the cooking of meat from uncastrated male pigs. Based on the intensity of this phenotype, the animals included in this study were classified into three groups: Low, Medium, and High boar taint.

Our two datasets originate from the 3 groups of boars, differentiated by their taints (Low - Medium - High taint), where one dataset contains Liver RNA-seq measurements and the other contains Methylation measurements. The RNA matrix consists of 75,375 genes measured across 9 boars (3 from each phenotypes), whereas the methylation matrix contains 100,000 methylation observations measured from the same 9 boars (3 from each phenotypes). For vertical integration using the MOFA framework, it is essential that the same biological samples are shared across all omics modalities. In our case, the nine boars were profiled at different molecular levels, allowing the integration of transcriptomic and methylation information from the same individuals.

```
##### load all the required files and arrange the files for MOFA Analysis
RNA_Seq_Data=read.csv("RNA_Seq_Liver.csv", row.names = 1)
head(RNA_Seq_Data)[1:5]
```

```
##                     X7107 X9138 X1306 X1113 X2111
## ENSSSCG00000000001     2     3     0     0     2
## ENSSSCG00000000002    20     9     2    23    34
## ENSSSCG00000000003  1207   945   350   488   708
## ENSSSCG00000000004    10     3     4     1     0
## ENSSSCG00000000005    30    51     8    35    52
## ENSSSCG00000000006   115    93   275    28    51
```

```
Methylation=read.csv("Epigenetic_matrix_filtered_100000.csv", row.names = 1)
rownames(Methylation)=Methylation$region
Methylation$X=NULL
Methylation$region =NULL

head(Methylation)
```

```
##                   X1105     X1137     X9312      X1113 X2111     X8308     X1306     X7107     X9138
## chr4_1126216  0.8750000 0.7195122 0.9629630 0.59677419  0.75 0.7884615 0.7894737 0.6410256 0.5555556
## chr18_2996989 0.0000000 0.4418605 0.0000000 1.00000000  0.00 0.3571429 0.3500000 0.0000000 0.3333333
## chr2_513273   0.0000000 0.0000000 0.0000000 0.00000000  0.00 0.0000000 0.0000000 0.0000000 0.0000000
## chr5_1441285  0.8888889 0.6796117 0.7857143 0.89583333  0.50 0.8888889 0.7800000 0.7173913 0.7500000
## chr3_44050407 0.0000000 1.0000000 0.0000000 0.00000000  1.00 0.0000000 1.0000000 1.0000000 1.0000000
## chr3_5000684  0.0000000 0.0000000 0.0000000 0.08333333  0.00 0.0000000 0.0000000 0.0000000 0.0000000
```

```
metadat_columns=read.csv("Metadata.csv")
head(metadat_columns)
```

```
##      X SampleID           Group BoarTaint
## 1 1326     1326 high boar taint         3
## 2 7106     7106 high boar taint         3
## 3 1331     1331 high boar taint         3
## 4 1334     1334 high boar taint         3
## 5 1338     1338 high boar taint         3
## 6 7107     7107 high boar taint         3
```

```
metadat_columns$X=NULL
```

---

3. Metadata filtering and sample alignment

The metadata file contains the identifiers of each boar together with its corresponding boar taint phenotype (Low, Medium, or High). In this step, the metadata are prepared to be compatible with the MOFA integration framework. One important concept during multi-omics integration is `shared samples measurements`. This means that the samples measurements from the individuals must be compatible across all other modalities. For example, if Dataset A contains the samples identifiers as ID1, ID2, ID3, then Dataset B and Dataset C must also contain the same identifier (ID1, ID2, ID3). Any mismatch in the sample naming would result in non-functioning data fusion, and consequently biasing the results outcome.

```
######################## Metadata Filtering ######################################
# Add prefix based on boar taint group
metadat_columns$SampleName <- ifelse(
  grepl("high", metadat_columns$Group, ignore.case = TRUE),
  paste0("H", metadat_columns$SampleID),
  ifelse(
    grepl("medium", metadat_columns$Group, ignore.case = TRUE),
    paste0("M", metadat_columns$SampleID),
    ifelse(
      grepl("low", metadat_columns$Group, ignore.case = TRUE),
      paste0("L", metadat_columns$SampleID),
      paste0("X", metadat_columns$SampleID)  # fallback if none match
    )
  )
)
metadat_columns$Group <- sub(" .*", "", metadat_columns$Group)



# Create the old and new column name mapping
metadat_columns$OldSampleName <- paste0("X", metadat_columns$SampleID)  # Original names
name_map <- setNames(metadat_columns$SampleName, metadat_columns$OldSampleName)  # Named vector: old ->
```

```r
# Rename methylation matrix columns
common_meth_cols <- intersect(colnames(Methylation), names(name_map))
colnames(Methylation)[match(common_meth_cols, colnames(Methylation))] <- name_map[common_meth_cols]
# Rename RNA-seq matrix columns
common_rna_cols <- intersect(colnames(RNA_Seq_Data), names(name_map))
colnames(RNA_Seq_Data)[match(common_rna_cols, colnames(RNA_Seq_Data))] <- name_map[common_rna_cols]
# Update metadata rownames to match new names
rownames(metadat_columns) <- metadat_columns$SampleName
# Filter metadata to keep only samples present in your omics data
omics_samples <- colnames(Methylation)
sample_metadata <- metadat_columns[metadat_columns$SampleName %in% omics_samples, ]


# Double-check that sample orders match
all(colnames(RNA_Seq_Data) == colnames(Methylation))
```

```
## [1] FALSE
```

```r
# Reorder columns of methylation matrix to match RNA-seq
meth_matrix <- Methylation[, colnames(RNA_Seq_Data)]
####### Check that they're now aligned
all(colnames(meth_matrix) == colnames(RNA_Seq_Data))  # should return TRUE
```

```
## [1] TRUE
```

```r
## Change the colnames of metadata
colnames(sample_metadata)[colnames(sample_metadata) == "SampleName"] <- "sample"
colnames(sample_metadata)[colnames(sample_metadata) == "Group"] <- "group"
```

---

4. Normalize and filter the RNA-seq data

Before integrating the transcriptomic data with any other data modality, the RNA-seq counts must be normalized to create a stable distribution. The raw count matrix are initially pre-processed by removing lowly expressed genes using `filterByExpr()`. Normalization factors are then estimated with `calcNormFactors()` for batch to batch estimation and false positive/negative results, and the counts are transformed into log2 Counts Per Million (CPM) values using a prior count to avoid undefined values produced by log(0) (mathematically undefined real number). The final processed RNA-seq data leaves with 9 boars and the top 5000 most variable genes.

```r
############ Normalize and filter for downstream analysis ######################


# Create DGEList object
dge <- DGEList(counts = RNA_Seq_Data)
keep <- filterByExpr(dge)
```

```
## Warning in filterByExpr.DGEList(dge): All samples appear to belong to the same group.
```

```r
dge <- dge[keep, , keep.lib.sizes=FALSE]
# Normalize with TMM
dge <- calcNormFactors(dge)
# Log2-CPM transformation
log_RNA <- cpm(dge, log = TRUE, prior.count = 1)
# Filter top 3000 most variable genes
gene_var <- apply(log_RNA, 1, var)
```

```r
top_genes <- names(sort(gene_var, decreasing = TRUE))[1:5000]
log_RNA_filtered <- log_RNA[top_genes, ]

dim(gene_var)
```

```
## NULL
```

```r
dim(log_RNA_filtered)
```

```
## [1] 5000    9
```

---

5. Transform and filter the methylation data

Before integrating the methylation data with any other data modalities, in the same process of RNA-seq, methylation data is subject to pre-processing. The data are first converted into a numeric matrix, after which the methylation -values are transformed into M-values using a logit transformatiion. This transformation is presented as log2 ratio transformation of the intensities methylated probes vs unmethylated probes. The reference of this transformation is in the following paper https://pmc.ncbi.nlm.nih.gov/articles/PMC3012676/. To improve data quality, methylation sites with more than 20% missing values are removed, and samples exceeding the same missingness threshold are also excluded. The final outcome of the methylation dataset is 5000 methylated regions in 9 candidate boars.

```r
### Methylation data
# Ensure beta values are in [0, 1]
Methylation[Methylation <= 0] <- 0
Methylation[Methylation >= 1] <- 1
summary(Methylation)
```

```
##      L1105             L1137             L9312             M1113             M2111             M8308
##  Min.   :0.00000   Min.   :0.00000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000
##  1st Qu.:0.00000   1st Qu.:0.01075   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000
##  Median :0.06667   Median :0.66216   Median :0.3333   Median :0.6316   Median :0.5789   Median :0.500
##  Mean   :0.40960   Mean   :0.51981   Mean   :0.4399   Mean   :0.5029   Mean   :0.4879   Mean   :0.473
##  3rd Qu.:1.00000   3rd Qu.:0.94545   3rd Qu.:0.9405   3rd Qu.:0.9091   3rd Qu.:0.9091   3rd Qu.:0.941
##  Max.   :1.00000   Max.   :1.00000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.000
##      H9138
##  Min.   :0.00000
##  1st Qu.:0.00000
##  Median :0.06977
##  Mean   :0.41240
##  3rd Qu.:1.00000
##  Max.   :1.00000
```

```r
# Convert data frame to numeric matrix
Methylation <- as.matrix(Methylation)
summary(Methylation)
```

```
##      L1105             L1137             L9312             M1113             M2111             M8308
##  Min.   :0.00000   Min.   :0.00000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000
##  1st Qu.:0.00000   1st Qu.:0.01075   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000
##  Median :0.06667   Median :0.66216   Median :0.3333   Median :0.6316   Median :0.5789   Median :0.500
##  Mean   :0.40960   Mean   :0.51981   Mean   :0.4399   Mean   :0.5029   Mean   :0.4879   Mean   :0.473
##  3rd Qu.:1.00000   3rd Qu.:0.94545   3rd Qu.:0.9405   3rd Qu.:0.9091   3rd Qu.:0.9091   3rd Qu.:0.941
```

```
##  Max.   :1.00000   Max.   :1.00000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.000
##      H9138
##  Min.   :0.00000
##  1st Qu.:0.00000
##  Median :0.06977
##  Mean   :0.41240
##  3rd Qu.:1.00000
##  Max.   :1.00000
```

```
storage.mode(Methylation) <- "numeric"  # make sure all values are numeric
# Convert to M-values
Mval <- log2(Methylation / (1 - Methylation))
# Replace Inf and -Inf with NA safely
Mval[is.infinite(Mval)] <- NA
class(Mval)
```

```
## [1] "matrix" "array"
```

```
# keep features with <20% NA
clean_meth <- Mval[apply(Mval, 1, function(x) sum(is.na(x)) < length(x) * 0.2), ]
clean_meth <- clean_meth[, colSums(is.na(clean_meth)) < nrow(clean_meth) * 0.2]

# Calculate variance per CpG (row), ignoring NAs
var_per_cpg <- apply(clean_meth, 1, var, na.rm = TRUE)
# Sort CpGs by variance decreasing and get top 5000 CpGs
top_CPGs <- names(sort(var_per_cpg, decreasing = TRUE))[1:5000]
# Subset methylation data to top variable CpGs
Meth_view <- clean_meth[top_CPGs, ]
# Check dimensions
dim(Meth_view)
```

```
## [1] 5000    9
```

```
# Confirm shared sample names
common_samples <- intersect(colnames(log_RNA_filtered), colnames(Meth_view))
```

---

6. Prepare the multi-omics data views

Before constructing the MOFA model, the RNA-seq and methylation datasets must contain the same biological samples in the same column order. The object common_samples, defined in the previous step, contains the sample identifiers shared by both omics datasets. The RNA-seq matrix is therefore subset using log_RNA_filtered[, common_samples], while the methylation matrix is subset using Meth_view[, common_samples]. This operation has two purposes. First, it removes any sample that is not present in both datasets. Second, it ensures that corresponding columns in the two matrices refer to the same biological individual. For example, the first column of RNA_view and the first column of Meth_view must represent the same sample. This alignment is essential because MOFA integrates information measured across the same set of samples.

The two processed datasets are then stored in the named list multi_omics_data. Each element of this list represents a separate omics view: - RNAseq contains the filtered and normalized RNA-seq data. - Methylation contains the filtered methylation M-values.

The function as.matrix() converts each object into a numeric matrix before it is supplied to MOFA. In both matrices, rows represent molecular features and columns represent samples. The RNA-seq matrix therefore contains genes in rows, whereas the methylation matrix contains CpG regions or methylation features in

rows. It is important to note that the datasets are not merged into a single dataframe. Instead, they remain as separate matrices within the same list. This preserves the identity of each omics modality and allows MOFA to model variation within and across the RNA-seq and methylation views.

```
# Subset and reorder both datasets to same sample order
RNA_view <- log_RNA_filtered[, common_samples]
Meth_view <- Meth_view[, common_samples]



# Build multi-omics data list (9 sample names in columns, rows have RNASeq in one object,
# similarly in a separate object another list with the CpG data)- They are NOT merged in a single objec
multi_omics_data <- list(
  RNAseq = as.matrix(RNA_view),
  Methylation = as.matrix(Meth_view)
)

head(multi_omics_data$RNAseq)
```

```
##                        H7107     H9138     H1306    M1113    M2111    M8308    L1105    L1137    L931
## ENSSSCG00000018700 3.577594 2.3951405 12.915918 5.818563 4.958452 3.532730 4.552157 3.095825 5.18135
## ENSSSCG00000020505 1.689962 1.0462637 11.095544 3.868868 3.111928 1.150505 2.262168 1.914087 2.87104
## ENSSSCG00000023855 1.527357 0.5622808 10.262853 3.290374 1.853726 1.602743 3.409988 1.832241 2.59485
## ENSSSCG00000022808 2.128574 1.9376806  7.564039 7.587772 9.016646 6.560589 8.588946 6.610569 7.09790
## ENSSSCG00000020439 7.407638 5.8595636 14.586316 8.186416 8.398337 6.757487 7.309273 6.711305 8.29647
## ENSSSCG00000019556 7.961373 6.8662736 15.150259 8.667477 8.015934 7.129994 7.994836 7.630179 8.64900
```

```
head(multi_omics_data$Methylation)
```

```
##                                 H7107      H9138       H1306       M1113      M2111      M8308      
## chrUn_NW_018084884v1_42920         NA  4.0000000 -0.53051472  4.66296501 -5.3923174 -6.7681843 -4.52
## chr13_205878354             0.3131579 -0.8479969  5.61470984  0.05062607  3.2479275 -6.4262648  4.70
## chr10_13233504              2.8524428  0.3219281 -6.50779464 -0.45943162  1.8930848  6.0000000      
## chr16_79245465              0.4909864  0.7369656  0.05889369  5.87036472         NA  0.2360674 -0.41
## chr9_6857263               -0.2410081 -0.3219281          NA  3.87446912  5.0000000  4.5849625  1.32
## chr5_88941641               0.1255309  1.7369656  4.33985000 -1.60880924 -0.8194278  2.5676845
```

---

7. Create, prepare, and train the MOFA model

Construction and training of the MOFA model

The two aligned omics views are used to construct the initial MOFA object with `create_mofa()`. The resulting object contains the RNA-seq and methylation matrices together with the sample-group assignments supplied through `groups = sample_metadata$group`. At this stage, the model has not yet been trained; the function only organizes the input data into the structure required by MOFA2. The `plot_data_overview()` function provides a visual summary of the input structure. It provides the number of datasets that we're currently willing to train, the sample numbers, MOFA is an unsupervised factor-analysis framework. It aims to represent the integrated datasets using a smaller number of latent factors. Each factor captures a source of variation that may be specific to one omics view or shared across multiple views. In this practical, the factors are inferred jointly from the RNA-seq and methylation data rather than being constructed to predict the Low, Medium, and High boar-taint classes.

The parameter `model_opts$num_factors <- 3` requests the estimation of three latent factors. These factors provide three independent dimensions through which the principal patterns of variability in the RNA-seq and

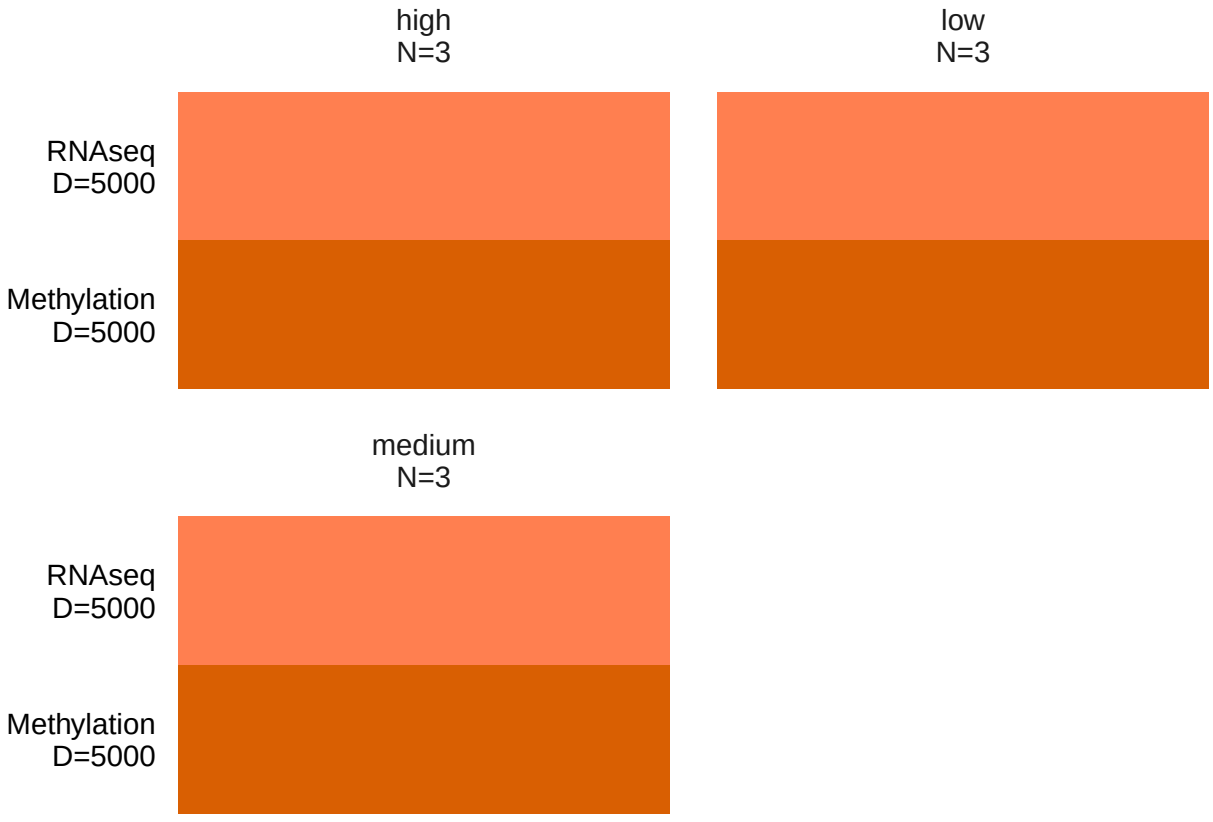
methylation datasets can be summarized. The number of factors is specified before training and should be interpreted in relation to the amount of variance explained by each factor during downstream analysis. The MOFA documentation states that there is no universally optimal number of factors because the appropriate choice depends on the objective, dimensions, and complexity of the analysis.

The `prepare_mofa()` function combines the selected data, model, and training options with the initial MOFA object. This preparation step checks and configures the object before model fitting. The sample metadata and groups are not supplied again because they were already incorporated when `create_mofa()` was called. Finally, `run_mofa()` trains the prepared model. The output file path is defined as `model.hdf5` within the current working directory. The HDF5 file stores the trained model and its learned parameters. Setting `use_basilisk = TRUE` instructs MOFA2 to use the Python environment managed through `basilisk`. For a clearer understanding of the integration, we invite you to look at the general framework in: `https://biofam.github.io/MOFA2/`.

```
# Create MOFA object with groups specified explicitly - it now access the list
mofa_object <- create_mofa(multi_omics_data, groups = sample_metadata$group)
```

```
## Creating MOFA object from a list of matrices (features as rows, sample as columns)...
```

```
plot_data_overview(mofa_object)
```




```
# Prepare options
data_opts <- get_default_data_options(mofa_object)
model_opts <- get_default_model_options(mofa_object)
train_opts <- get_default_training_options(mofa_object)
train_opts$seed <- 42  # for reproducibility
train_opts$maxiter <- 1000
model_opts$num_factors <- 3
```

```
# Prepare the model (NO samples_metadata or groups arguments here)
mofa_object <- prepare_mofa(
  object = mofa_object,
  data_options = data_opts,
  model_options = model_opts,
  training_options = train_opts
)
```

```
## Warning in prepare_mofa(object = mofa_object, data_options = data_opts, : Some group(s) have less tha
## learn meaningful factors for these group(s)...

##
## # Multi-group mode requested.

##
## This is an advanced option, if this is the first time that you are running MOFA, we suggest that you

##
##  - The aim of the multi-group framework is to identify the sources of variability *within* the groups

##
##  - It is important to account for the group effect before selecting highly variable features (HVFs).

## Checking data options...

## Checking training options...

## Checking model options...

## Warning in prepare_mofa(object = mofa_object, data_options = data_opts, : The total number of samples
##     Try to reduce the number of factors to obtain meaningful results. It should not exceed ~0.
```

```
# Run MOFA
outfile = file.path(getwd(),"model.hdf5")
mofa_trained <- run_mofa(mofa_object,outfile, use_basilisk = TRUE)
```

```
## Warning: Output file /home/adam/Desktop/China_Workshop/training_multiomics/mofa/model.hdf5 already ex

## Connecting to the mofapy2 package using basilisk.
##     Set 'use_basilisk' to FALSE if you prefer to manually set the python binary using 'reticulate'.

##
##          ############################################################
##          ###           __  __  ____  ______                       ###
##          ###          |  \/  |/ __ \|  ____/\    _                ###
##          ###          | \  / | |  | | |__ /  \ _| |_              ###
##          ###          | |\/| | |  | |  __/ /\ \_   _|             ###
##          ###          | |  | | |__| | | / ____ \|_|               ###
##          ###          |_|  |_|\____/|_|/_/    \_\                ###
##          ###                                                      ###
##          ############################################################
##
##
##
## use_float32 set to True: replacing float64 arrays by float32 arrays to speed up computations...
##
## Successfully loaded view='RNAseq' group='high' with N=3 samples and D=5000 features...
## Successfully loaded view='RNAseq' group='low' with N=3 samples and D=5000 features...
## Successfully loaded view='RNAseq' group='medium' with N=3 samples and D=5000 features...
```

```
## Successfully loaded view='Methylation' group='high' with N=3 samples and D=5000 features...
## Successfully loaded view='Methylation' group='low' with N=3 samples and D=5000 features...
## Successfully loaded view='Methylation' group='medium' with N=3 samples and D=5000 features...
##
##
## Model options:
## - Automatic Relevance Determination prior on the factors: True
## - Automatic Relevance Determination prior on the weights: True
## - Spike-and-slab prior on the factors: False
## - Spike-and-slab prior on the weights: False
## Likelihoods:
## - View 0 (RNAseq): gaussian
## - View 1 (Methylation): gaussian
##
##
##
## Warning: some group(s) have less than 15 samples, MOFA won't be able to learn meaningful factors for
##
##
##
## ######################################
## ## Training the model with seed 42 ##
## ######################################
##
##
## ELBO before training: -492753.71
##
## Iteration 1: time=0.01, ELBO=-277683.08, deltaELBO=215070.626 (43.64667840%), Factors=3
## Iteration 2: time=0.00, Factors=3
## Iteration 3: time=0.00, Factors=3
## Iteration 4: time=0.00, Factors=3
## Iteration 5: time=0.00, Factors=3
## Iteration 6: time=0.01, ELBO=-254721.18, deltaELBO=22961.906 (4.65991546%), Factors=3
## Iteration 7: time=0.00, Factors=3
## Iteration 8: time=0.00, Factors=3
## Iteration 9: time=0.00, Factors=3
## Iteration 10: time=0.00, Factors=3
## Iteration 11: time=0.01, ELBO=-250994.69, deltaELBO=3726.485 (0.75625717%), Factors=3
## Iteration 12: time=0.00, Factors=3
## Iteration 13: time=0.00, Factors=3
## Iteration 14: time=0.00, Factors=3
## Iteration 15: time=0.00, Factors=3
## Iteration 16: time=0.01, ELBO=-248910.86, deltaELBO=2083.829 (0.42289462%), Factors=3
## Iteration 17: time=0.00, Factors=3
## Iteration 18: time=0.00, Factors=3
## Iteration 19: time=0.00, Factors=3
## Iteration 20: time=0.00, Factors=3
## Iteration 21: time=0.01, ELBO=-247299.44, deltaELBO=1611.419 (0.32702314%), Factors=3
## Iteration 22: time=0.00, Factors=3
## Iteration 23: time=0.00, Factors=3
## Iteration 24: time=0.00, Factors=3
## Iteration 25: time=0.00, Factors=3
## Iteration 26: time=0.00, ELBO=-246162.87, deltaELBO=1136.572 (0.23065718%), Factors=3
## Iteration 27: time=0.00, Factors=3
```

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## Iteration 28: time=0.00, Factors=3
## Iteration 29: time=0.00, Factors=3
## Iteration 30: time=0.00, Factors=3
## Iteration 31: time=0.01, ELBO=-245431.97, deltaELBO=730.897 (0.14832917%), Factors=3
## Iteration 32: time=0.00, Factors=3
## Iteration 33: time=0.00, Factors=3
## Iteration 34: time=0.00, Factors=3
## Iteration 35: time=0.00, Factors=3
## Iteration 36: time=0.01, ELBO=-244920.18, deltaELBO=511.797 (0.10386469%), Factors=3
## Iteration 37: time=0.00, Factors=3
## Iteration 38: time=0.00, Factors=3
## Iteration 39: time=0.00, Factors=3
## Iteration 40: time=0.00, Factors=3
## Iteration 41: time=0.01, ELBO=-244522.71, deltaELBO=397.470 (0.08066295%), Factors=3
## Iteration 42: time=0.00, Factors=3
## Iteration 43: time=0.00, Factors=3
## Iteration 44: time=0.00, Factors=3
## Iteration 45: time=0.00, Factors=3
## Iteration 46: time=0.01, ELBO=-244205.97, deltaELBO=316.741 (0.06427972%), Factors=3
## Iteration 47: time=0.00, Factors=3
## Iteration 48: time=0.00, Factors=3
## Iteration 49: time=0.00, Factors=3
## Iteration 50: time=0.00, Factors=3
## Iteration 51: time=0.01, ELBO=-243933.48, deltaELBO=272.483 (0.05529802%), Factors=3
## Iteration 52: time=0.00, Factors=3
## Iteration 53: time=0.00, Factors=3
## Iteration 54: time=0.00, Factors=3
## Iteration 55: time=0.01, Factors=3
## Iteration 56: time=0.02, ELBO=-243702.97, deltaELBO=230.511 (0.04678010%), Factors=3
## Iteration 57: time=0.01, Factors=3
## Iteration 58: time=0.01, Factors=3
## Iteration 59: time=0.00, Factors=3
## Iteration 60: time=0.00, Factors=3
## Iteration 61: time=0.01, ELBO=-243505.38, deltaELBO=197.591 (0.04009927%), Factors=3
## Iteration 62: time=0.01, Factors=3
## Iteration 63: time=0.00, Factors=3
## Iteration 64: time=0.00, Factors=3
## Iteration 65: time=0.00, Factors=3
## Iteration 66: time=0.01, ELBO=-243330.99, deltaELBO=174.389 (0.03539062%), Factors=3
## Iteration 67: time=0.00, Factors=3
## Iteration 68: time=0.00, Factors=3
## Iteration 69: time=0.00, Factors=3
## Iteration 70: time=0.00, Factors=3
## Iteration 71: time=0.01, ELBO=-243188.07, deltaELBO=142.927 (0.02900579%), Factors=3
## Iteration 72: time=0.00, Factors=3
## Iteration 73: time=0.00, Factors=3
## Iteration 74: time=0.00, Factors=3
## Iteration 75: time=0.00, Factors=3
## Iteration 76: time=0.01, ELBO=-243074.87, deltaELBO=113.198 (0.02297248%), Factors=3
## Iteration 77: time=0.00, Factors=3
## Iteration 78: time=0.00, Factors=3
## Iteration 79: time=0.00, Factors=3
## Iteration 80: time=0.00, Factors=3
## Iteration 81: time=0.01, ELBO=-242970.67, deltaELBO=104.196 (0.02114573%), Factors=3
```

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## Iteration 82: time=0.00, Factors=3
## Iteration 83: time=0.00, Factors=3
## Iteration 84: time=0.00, Factors=3
## Iteration 85: time=0.00, Factors=3
## Iteration 86: time=0.01, ELBO=-242879.33, deltaELBO=91.343 (0.01853724%), Factors=3
## Iteration 87: time=0.00, Factors=3
## Iteration 88: time=0.00, Factors=3
## Iteration 89: time=0.00, Factors=3
## Iteration 90: time=0.00, Factors=3
## Iteration 91: time=0.01, ELBO=-242795.22, deltaELBO=84.110 (0.01706947%), Factors=3
## Iteration 92: time=0.00, Factors=3
## Iteration 93: time=0.00, Factors=3
## Iteration 94: time=0.00, Factors=3
## Iteration 95: time=0.00, Factors=3
## Iteration 96: time=0.01, ELBO=-242722.59, deltaELBO=72.626 (0.01473872%), Factors=3
## Iteration 97: time=0.00, Factors=3
## Iteration 98: time=0.00, Factors=3
## Iteration 99: time=0.00, Factors=3
## Iteration 100: time=0.00, Factors=3
## Iteration 101: time=0.01, ELBO=-242656.30, deltaELBO=66.289 (0.01345274%), Factors=3
## Iteration 102: time=0.00, Factors=3
## Iteration 103: time=0.00, Factors=3
## Iteration 104: time=0.00, Factors=3
## Iteration 105: time=0.00, Factors=3
## Iteration 106: time=0.01, ELBO=-242597.91, deltaELBO=58.390 (0.01184963%), Factors=3
## Iteration 107: time=0.00, Factors=3
## Iteration 108: time=0.00, Factors=3
## Iteration 109: time=0.00, Factors=3
## Iteration 110: time=0.00, Factors=3
## Iteration 111: time=0.01, ELBO=-242544.47, deltaELBO=53.447 (0.01084669%), Factors=3
## Iteration 112: time=0.00, Factors=3
## Iteration 113: time=0.00, Factors=3
## Iteration 114: time=0.00, Factors=3
## Iteration 115: time=0.00, Factors=3
## Iteration 116: time=0.01, ELBO=-242495.77, deltaELBO=48.701 (0.00988335%), Factors=3
## Iteration 117: time=0.00, Factors=3
## Iteration 118: time=0.00, Factors=3
## Iteration 119: time=0.00, Factors=3
## Iteration 120: time=0.00, Factors=3
## Iteration 121: time=0.01, ELBO=-242453.04, deltaELBO=42.722 (0.00867012%), Factors=3
## Iteration 122: time=0.00, Factors=3
## Iteration 123: time=0.00, Factors=3
## Iteration 124: time=0.00, Factors=3
## Iteration 125: time=0.00, Factors=3
## Iteration 126: time=0.01, ELBO=-242413.86, deltaELBO=39.186 (0.00795237%), Factors=3
## Iteration 127: time=0.00, Factors=3
## Iteration 128: time=0.00, Factors=3
## Iteration 129: time=0.00, Factors=3
## Iteration 130: time=0.00, Factors=3
## Iteration 131: time=0.01, ELBO=-242378.30, deltaELBO=35.558 (0.00721619%), Factors=3
## Iteration 132: time=0.00, Factors=3
## Iteration 133: time=0.00, Factors=3
## Iteration 134: time=0.00, Factors=3
## Iteration 135: time=0.00, Factors=3
```

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## Iteration 136: time=0.01, ELBO=-242341.76, deltaELBO=36.536 (0.00741460%), Factors=3
## Iteration 137: time=0.00, Factors=3
## Iteration 138: time=0.00, Factors=3
## Iteration 139: time=0.00, Factors=3
## Iteration 140: time=0.00, Factors=3
## Iteration 141: time=0.01, ELBO=-242299.25, deltaELBO=42.519 (0.00862893%), Factors=3
## Iteration 142: time=0.00, Factors=3
## Iteration 143: time=0.00, Factors=3
## Iteration 144: time=0.00, Factors=3
## Iteration 145: time=0.00, Factors=3
## Iteration 146: time=0.01, ELBO=-242261.54, deltaELBO=37.707 (0.00765235%), Factors=3
## Iteration 147: time=0.00, Factors=3
## Iteration 148: time=0.00, Factors=3
## Iteration 149: time=0.00, Factors=3
## Iteration 150: time=0.00, Factors=3
## Iteration 151: time=0.01, ELBO=-242228.95, deltaELBO=32.584 (0.00661264%), Factors=3
## Iteration 152: time=0.01, Factors=3
## Iteration 153: time=0.00, Factors=3
## Iteration 154: time=0.00, Factors=3
## Iteration 155: time=0.00, Factors=3
## Iteration 156: time=0.01, ELBO=-242195.08, deltaELBO=33.875 (0.00687463%), Factors=3
## Iteration 157: time=0.00, Factors=3
## Iteration 158: time=0.00, Factors=3
## Iteration 159: time=0.00, Factors=3
## Iteration 160: time=0.00, Factors=3
## Iteration 161: time=0.01, ELBO=-242164.06, deltaELBO=31.024 (0.00629605%), Factors=3
## Iteration 162: time=0.00, Factors=3
## Iteration 163: time=0.00, Factors=3
## Iteration 164: time=0.00, Factors=3
## Iteration 165: time=0.00, Factors=3
## Iteration 166: time=0.01, ELBO=-242135.62, deltaELBO=28.437 (0.00577099%), Factors=3
## Iteration 167: time=0.00, Factors=3
## Iteration 168: time=0.00, Factors=3
## Iteration 169: time=0.00, Factors=3
## Iteration 170: time=0.00, Factors=3
## Iteration 171: time=0.01, ELBO=-242106.67, deltaELBO=28.951 (0.00587545%), Factors=3
## Iteration 172: time=0.00, Factors=3
## Iteration 173: time=0.00, Factors=3
## Iteration 174: time=0.00, Factors=3
## Iteration 175: time=0.03, Factors=3
## Iteration 176: time=0.02, ELBO=-242073.65, deltaELBO=33.014 (0.00669996%), Factors=3
## Iteration 177: time=0.00, Factors=3
## Iteration 178: time=0.00, Factors=3
## Iteration 179: time=0.00, Factors=3
## Iteration 180: time=0.00, Factors=3
## Iteration 181: time=0.01, ELBO=-242049.25, deltaELBO=24.403 (0.00495238%), Factors=3
## Iteration 182: time=0.00, Factors=3
## Iteration 183: time=0.00, Factors=3
## Iteration 184: time=0.00, Factors=3
## Iteration 185: time=0.00, Factors=3
## Iteration 186: time=0.01, ELBO=-242030.49, deltaELBO=18.760 (0.00380712%), Factors=3
## Iteration 187: time=0.00, Factors=3
## Iteration 188: time=0.00, Factors=3
## Iteration 189: time=0.00, Factors=3
```

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## Iteration 190: time=0.00, Factors=3
## Iteration 191: time=0.01, ELBO=-242012.29, deltaELBO=18.196 (0.00369269%), Factors=3
## Iteration 192: time=0.00, Factors=3
## Iteration 193: time=0.00, Factors=3
## Iteration 194: time=0.00, Factors=3
## Iteration 195: time=0.00, Factors=3
## Iteration 196: time=0.01, ELBO=-241991.13, deltaELBO=21.162 (0.00429471%), Factors=3
## Iteration 197: time=0.00, Factors=3
## Iteration 198: time=0.00, Factors=3
## Iteration 199: time=0.00, Factors=3
## Iteration 200: time=0.00, Factors=3
## Iteration 201: time=0.01, ELBO=-241973.74, deltaELBO=17.390 (0.00352909%), Factors=3
## Iteration 202: time=0.00, Factors=3
## Iteration 203: time=0.00, Factors=3
## Iteration 204: time=0.00, Factors=3
## Iteration 205: time=0.00, Factors=3
## Iteration 206: time=0.01, ELBO=-241959.10, deltaELBO=14.637 (0.00297051%), Factors=3
## Iteration 207: time=0.00, Factors=3
## Iteration 208: time=0.00, Factors=3
## Iteration 209: time=0.00, Factors=3
## Iteration 210: time=0.00, Factors=3
## Iteration 211: time=0.01, ELBO=-241944.33, deltaELBO=14.774 (0.00299833%), Factors=3
## Iteration 212: time=0.00, Factors=3
## Iteration 213: time=0.00, Factors=3
## Iteration 214: time=0.00, Factors=3
## Iteration 215: time=0.00, Factors=3
## Iteration 216: time=0.01, ELBO=-241928.04, deltaELBO=16.288 (0.00330549%), Factors=3
## Iteration 217: time=0.00, Factors=3
## Iteration 218: time=0.00, Factors=3
## Iteration 219: time=0.00, Factors=3
## Iteration 220: time=0.00, Factors=3
## Iteration 221: time=0.01, ELBO=-241911.56, deltaELBO=16.480 (0.00334453%), Factors=3
## Iteration 222: time=0.00, Factors=3
## Iteration 223: time=0.00, Factors=3
## Iteration 224: time=0.00, Factors=3
## Iteration 225: time=0.00, Factors=3
## Iteration 226: time=0.01, ELBO=-241896.06, deltaELBO=15.504 (0.00314636%), Factors=3
## Iteration 227: time=0.00, Factors=3
## Iteration 228: time=0.00, Factors=3
## Iteration 229: time=0.00, Factors=3
## Iteration 230: time=0.00, Factors=3
## Iteration 231: time=0.01, ELBO=-241881.17, deltaELBO=14.883 (0.00302042%), Factors=3
## Iteration 232: time=0.00, Factors=3
## Iteration 233: time=0.00, Factors=3
## Iteration 234: time=0.00, Factors=3
## Iteration 235: time=0.00, Factors=3
## Iteration 236: time=0.01, ELBO=-241867.03, deltaELBO=14.141 (0.00286989%), Factors=3
## Iteration 237: time=0.00, Factors=3
## Iteration 238: time=0.00, Factors=3
## Iteration 239: time=0.00, Factors=3
## Iteration 240: time=0.00, Factors=3
## Iteration 241: time=0.01, ELBO=-241849.46, deltaELBO=17.577 (0.00356710%), Factors=3
## Iteration 242: time=0.00, Factors=3
## Iteration 243: time=0.00, Factors=3
```

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## Iteration 244: time=0.00, Factors=3
## Iteration 245: time=0.00, Factors=3
## Iteration 246: time=0.01, ELBO=-241831.64, deltaELBO=17.816 (0.00361552%), Factors=3
## Iteration 247: time=0.00, Factors=3
## Iteration 248: time=0.00, Factors=3
## Iteration 249: time=0.00, Factors=3
## Iteration 250: time=0.00, Factors=3
## Iteration 251: time=0.01, ELBO=-241819.97, deltaELBO=11.668 (0.00236797%), Factors=3
## Iteration 252: time=0.00, Factors=3
## Iteration 253: time=0.00, Factors=3
## Iteration 254: time=0.00, Factors=3
## Iteration 255: time=0.00, Factors=3
## Iteration 256: time=0.01, ELBO=-241808.55, deltaELBO=11.427 (0.00231899%), Factors=3
## Iteration 257: time=0.00, Factors=3
## Iteration 258: time=0.00, Factors=3
## Iteration 259: time=0.00, Factors=3
## Iteration 260: time=0.00, Factors=3
## Iteration 261: time=0.01, ELBO=-241796.42, deltaELBO=12.130 (0.00246176%), Factors=3
## Iteration 262: time=0.00, Factors=3
## Iteration 263: time=0.00, Factors=3
## Iteration 264: time=0.00, Factors=3
## Iteration 265: time=0.00, Factors=3
## Iteration 266: time=0.01, ELBO=-241784.09, deltaELBO=12.321 (0.00250034%), Factors=3
## Iteration 267: time=0.00, Factors=3
## Iteration 268: time=0.00, Factors=3
## Iteration 269: time=0.00, Factors=3
## Iteration 270: time=0.00, Factors=3
## Iteration 271: time=0.01, ELBO=-241774.07, deltaELBO=10.028 (0.00203514%), Factors=3
## Iteration 272: time=0.00, Factors=3
## Iteration 273: time=0.00, Factors=3
## Iteration 274: time=0.00, Factors=3
## Iteration 275: time=0.00, Factors=3
## Iteration 276: time=0.01, ELBO=-241760.95, deltaELBO=13.121 (0.00266270%), Factors=3
## Iteration 277: time=0.00, Factors=3
## Iteration 278: time=0.00, Factors=3
## Iteration 279: time=0.00, Factors=3
## Iteration 280: time=0.00, Factors=3
## Iteration 281: time=0.01, ELBO=-241745.25, deltaELBO=15.695 (0.00318510%), Factors=3
## Iteration 282: time=0.00, Factors=3
## Iteration 283: time=0.00, Factors=3
## Iteration 284: time=0.00, Factors=3
## Iteration 285: time=0.00, Factors=3
## Iteration 286: time=0.01, ELBO=-241731.89, deltaELBO=13.364 (0.00271217%), Factors=3
## Iteration 287: time=0.00, Factors=3
## Iteration 288: time=0.00, Factors=3
## Iteration 289: time=0.00, Factors=3
## Iteration 290: time=0.00, Factors=3
## Iteration 291: time=0.01, ELBO=-241720.92, deltaELBO=10.966 (0.00222536%), Factors=3
## Iteration 292: time=0.01, Factors=3
## Iteration 293: time=0.03, Factors=3
## Iteration 294: time=0.02, Factors=3
## Iteration 295: time=0.00, Factors=3
## Iteration 296: time=0.01, ELBO=-241710.04, deltaELBO=10.880 (0.00220799%), Factors=3
## Iteration 297: time=0.00, Factors=3
```

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## Iteration 298: time=0.00, Factors=3
## Iteration 299: time=0.00, Factors=3
## Iteration 300: time=0.00, Factors=3
## Iteration 301: time=0.01, ELBO=-241702.02, deltaELBO=8.018 (0.00162719%), Factors=3
## Iteration 302: time=0.01, Factors=3
## Iteration 303: time=0.00, Factors=3
## Iteration 304: time=0.00, Factors=3
## Iteration 305: time=0.00, Factors=3
## Iteration 306: time=0.01, ELBO=-241693.42, deltaELBO=8.604 (0.00174618%), Factors=3
## Iteration 307: time=0.00, Factors=3
## Iteration 308: time=0.00, Factors=3
## Iteration 309: time=0.00, Factors=3
## Iteration 310: time=0.00, Factors=3
## Iteration 311: time=0.01, ELBO=-241682.89, deltaELBO=10.532 (0.00213730%), Factors=3
## Iteration 312: time=0.00, Factors=3
## Iteration 313: time=0.00, Factors=3
## Iteration 314: time=0.00, Factors=3
## Iteration 315: time=0.00, Factors=3
## Iteration 316: time=0.01, ELBO=-241672.86, deltaELBO=10.032 (0.00203591%), Factors=3
## Iteration 317: time=0.00, Factors=3
## Iteration 318: time=0.00, Factors=3
## Iteration 319: time=0.00, Factors=3
## Iteration 320: time=0.00, Factors=3
## Iteration 321: time=0.01, ELBO=-241661.02, deltaELBO=11.831 (0.00240100%), Factors=3
## Iteration 322: time=0.00, Factors=3
## Iteration 323: time=0.00, Factors=3
## Iteration 324: time=0.00, Factors=3
## Iteration 325: time=0.00, Factors=3
## Iteration 326: time=0.01, ELBO=-241652.13, deltaELBO=8.892 (0.00180454%), Factors=3
## Iteration 327: time=0.00, Factors=3
## Iteration 328: time=0.00, Factors=3
## Iteration 329: time=0.00, Factors=3
## Iteration 330: time=0.00, Factors=3
## Iteration 331: time=0.01, ELBO=-241644.88, deltaELBO=7.253 (0.00147185%), Factors=3
## Iteration 332: time=0.00, Factors=3
## Iteration 333: time=0.00, Factors=3
## Iteration 334: time=0.00, Factors=3
## Iteration 335: time=0.00, Factors=3
## Iteration 336: time=0.01, ELBO=-241637.57, deltaELBO=7.305 (0.00148247%), Factors=3
## Iteration 337: time=0.00, Factors=3
## Iteration 338: time=0.00, Factors=3
## Iteration 339: time=0.00, Factors=3
## Iteration 340: time=0.00, Factors=3
## Iteration 341: time=0.01, ELBO=-241630.24, deltaELBO=7.340 (0.00148953%), Factors=3
## Iteration 342: time=0.00, Factors=3
## Iteration 343: time=0.00, Factors=3
## Iteration 344: time=0.00, Factors=3
## Iteration 345: time=0.00, Factors=3
## Iteration 346: time=0.01, ELBO=-241621.27, deltaELBO=8.970 (0.00182037%), Factors=3
## Iteration 347: time=0.00, Factors=3
## Iteration 348: time=0.00, Factors=3
## Iteration 349: time=0.00, Factors=3
## Iteration 350: time=0.00, Factors=3
## Iteration 351: time=0.01, ELBO=-241611.10, deltaELBO=10.165 (0.00206298%), Factors=3
```

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## Iteration 352: time=0.00, Factors=3
## Iteration 353: time=0.01, Factors=3
## Iteration 354: time=0.00, Factors=3
## Iteration 355: time=0.00, Factors=3
## Iteration 356: time=0.01, ELBO=-241601.67, deltaELBO=9.434 (0.00191457%), Factors=3
## Iteration 357: time=0.00, Factors=3
## Iteration 358: time=0.00, Factors=3
## Iteration 359: time=0.01, Factors=3
## Iteration 360: time=0.00, Factors=3
## Iteration 361: time=0.01, ELBO=-241594.25, deltaELBO=7.413 (0.00150445%), Factors=3
## Iteration 362: time=0.00, Factors=3
## Iteration 363: time=0.00, Factors=3
## Iteration 364: time=0.00, Factors=3
## Iteration 365: time=0.00, Factors=3
## Iteration 366: time=0.01, ELBO=-241584.77, deltaELBO=9.481 (0.00192414%), Factors=3
## Iteration 367: time=0.00, Factors=3
## Iteration 368: time=0.00, Factors=3
## Iteration 369: time=0.00, Factors=3
## Iteration 370: time=0.00, Factors=3
## Iteration 371: time=0.01, ELBO=-241573.33, deltaELBO=11.438 (0.00232119%), Factors=3
## Iteration 372: time=0.00, Factors=3
## Iteration 373: time=0.00, Factors=3
## Iteration 374: time=0.00, Factors=3
## Iteration 375: time=0.00, Factors=3
## Iteration 376: time=0.01, ELBO=-241564.82, deltaELBO=8.509 (0.00172676%), Factors=3
## Iteration 377: time=0.00, Factors=3
## Iteration 378: time=0.00, Factors=3
## Iteration 379: time=0.00, Factors=3
## Iteration 380: time=0.00, Factors=3
## Iteration 381: time=0.01, ELBO=-241557.08, deltaELBO=7.747 (0.00157217%), Factors=3
## Iteration 382: time=0.00, Factors=3
## Iteration 383: time=0.00, Factors=3
## Iteration 384: time=0.00, Factors=3
## Iteration 385: time=0.00, Factors=3
## Iteration 386: time=0.01, ELBO=-241550.19, deltaELBO=6.892 (0.00139875%), Factors=3
## Iteration 387: time=0.00, Factors=3
## Iteration 388: time=0.00, Factors=3
## Iteration 389: time=0.00, Factors=3
## Iteration 390: time=0.00, Factors=3
## Iteration 391: time=0.01, ELBO=-241542.57, deltaELBO=7.613 (0.00154491%), Factors=3
## Iteration 392: time=0.00, Factors=3
## Iteration 393: time=0.00, Factors=3
## Iteration 394: time=0.00, Factors=3
## Iteration 395: time=0.00, Factors=3
## Iteration 396: time=0.01, ELBO=-241533.76, deltaELBO=8.809 (0.00178781%), Factors=3
## Iteration 397: time=0.00, Factors=3
## Iteration 398: time=0.00, Factors=3
## Iteration 399: time=0.00, Factors=3
## Iteration 400: time=0.00, Factors=3
## Iteration 401: time=0.01, ELBO=-241525.40, deltaELBO=8.367 (0.00169792%), Factors=3
## Iteration 402: time=0.00, Factors=3
## Iteration 403: time=0.01, Factors=3
## Iteration 404: time=0.02, Factors=3
## Iteration 405: time=0.01, Factors=3
```

```
## Iteration 406: time=0.02, ELBO=-241517.97, deltaELBO=7.428 (0.00150735%), Factors=3
## Iteration 407: time=0.00, Factors=3
## Iteration 408: time=0.00, Factors=3
## Iteration 409: time=0.00, Factors=3
## Iteration 410: time=0.00, Factors=3
## Iteration 411: time=0.01, ELBO=-241510.45, deltaELBO=7.519 (0.00152594%), Factors=3
## Iteration 412: time=0.00, Factors=3
## Iteration 413: time=0.00, Factors=3
## Iteration 414: time=0.00, Factors=3
## Iteration 415: time=0.00, Factors=3
## Iteration 416: time=0.01, ELBO=-241501.86, deltaELBO=8.586 (0.00174240%), Factors=3
## Iteration 417: time=0.00, Factors=3
## Iteration 418: time=0.00, Factors=3
## Iteration 419: time=0.00, Factors=3
## Iteration 420: time=0.00, Factors=3
## Iteration 421: time=0.01, ELBO=-241491.89, deltaELBO=9.978 (0.00202488%), Factors=3
## Iteration 422: time=0.00, Factors=3
## Iteration 423: time=0.00, Factors=3
## Iteration 424: time=0.00, Factors=3
## Iteration 425: time=0.00, Factors=3
## Iteration 426: time=0.01, ELBO=-241485.17, deltaELBO=6.714 (0.00136245%), Factors=3
## Iteration 427: time=0.00, Factors=3
## Iteration 428: time=0.00, Factors=3
## Iteration 429: time=0.00, Factors=3
## Iteration 430: time=0.00, Factors=3
## Iteration 431: time=0.01, ELBO=-241477.00, deltaELBO=8.177 (0.00165937%), Factors=3
## Iteration 432: time=0.00, Factors=3
## Iteration 433: time=0.00, Factors=3
## Iteration 434: time=0.00, Factors=3
## Iteration 435: time=0.00, Factors=3
## Iteration 436: time=0.01, ELBO=-241467.31, deltaELBO=9.689 (0.00196631%), Factors=3
## Iteration 437: time=0.00, Factors=3
## Iteration 438: time=0.00, Factors=3
## Iteration 439: time=0.00, Factors=3
## Iteration 440: time=0.00, Factors=3
## Iteration 441: time=0.01, ELBO=-241461.40, deltaELBO=5.907 (0.00119887%), Factors=3
## Iteration 442: time=0.00, Factors=3
## Iteration 443: time=0.00, Factors=3
## Iteration 444: time=0.00, Factors=3
## Iteration 445: time=0.00, Factors=3
## Iteration 446: time=0.01, ELBO=-241457.90, deltaELBO=3.500 (0.00071030%), Factors=3
## Iteration 447: time=0.00, Factors=3
## Iteration 448: time=0.01, Factors=3
## Iteration 449: time=0.00, Factors=3
## Iteration 450: time=0.00, Factors=3
## Iteration 451: time=0.01, ELBO=-241455.05, deltaELBO=2.847 (0.00057783%), Factors=3
## Iteration 452: time=0.00, Factors=3
## Iteration 453: time=0.00, Factors=3
## Iteration 454: time=0.00, Factors=3
## Iteration 455: time=0.00, Factors=3
## Iteration 456: time=0.01, ELBO=-241452.08, deltaELBO=2.970 (0.00060264%), Factors=3
## Iteration 457: time=0.00, Factors=3
## Iteration 458: time=0.00, Factors=3
## Iteration 459: time=0.00, Factors=3
```

```
## Iteration 460: time=0.00, Factors=3
## Iteration 461: time=0.01, ELBO=-241447.71, deltaELBO=4.374 (0.00088761%), Factors=3
## Iteration 462: time=0.00, Factors=3
## Iteration 463: time=0.00, Factors=3
## Iteration 464: time=0.00, Factors=3
## Iteration 465: time=0.00, Factors=3
## Iteration 466: time=0.01, ELBO=-241439.81, deltaELBO=7.898 (0.00160279%), Factors=3
## Iteration 467: time=0.00, Factors=3
## Iteration 468: time=0.00, Factors=3
## Iteration 469: time=0.00, Factors=3
## Iteration 470: time=0.00, Factors=3
## Iteration 471: time=0.01, ELBO=-241432.83, deltaELBO=6.977 (0.00141597%), Factors=3
## Iteration 472: time=0.00, Factors=3
## Iteration 473: time=0.00, Factors=3
## Iteration 474: time=0.00, Factors=3
## Iteration 475: time=0.00, Factors=3
## Iteration 476: time=0.01, ELBO=-241425.35, deltaELBO=7.482 (0.00151847%), Factors=3
## Iteration 477: time=0.00, Factors=3
## Iteration 478: time=0.00, Factors=3
## Iteration 479: time=0.00, Factors=3
## Iteration 480: time=0.00, Factors=3
## Iteration 481: time=0.01, ELBO=-241418.02, deltaELBO=7.330 (0.00148757%), Factors=3
## Iteration 482: time=0.00, Factors=3
## Iteration 483: time=0.00, Factors=3
## Iteration 484: time=0.00, Factors=3
## Iteration 485: time=0.00, Factors=3
## Iteration 486: time=0.01, ELBO=-241412.85, deltaELBO=5.170 (0.00104921%), Factors=3
## Iteration 487: time=0.00, Factors=3
## Iteration 488: time=0.00, Factors=3
## Iteration 489: time=0.00, Factors=3
## Iteration 490: time=0.00, Factors=3
## Iteration 491: time=0.01, ELBO=-241409.20, deltaELBO=3.652 (0.00074114%), Factors=3
## Iteration 492: time=0.02, Factors=3
## Iteration 493: time=0.02, Factors=3
## Iteration 494: time=0.02, Factors=3
## Iteration 495: time=0.02, Factors=3
## Iteration 496: time=0.03, ELBO=-241404.50, deltaELBO=4.696 (0.00095311%), Factors=3
## Iteration 497: time=0.02, Factors=3
## Iteration 498: time=0.02, Factors=3
## Iteration 499: time=0.03, Factors=3
## Iteration 500: time=0.00, Factors=3
## Iteration 501: time=0.01, ELBO=-241399.66, deltaELBO=4.847 (0.00098366%), Factors=3
## Iteration 502: time=0.00, Factors=3
## Iteration 503: time=0.00, Factors=3
## Iteration 504: time=0.00, Factors=3
## Iteration 505: time=0.00, Factors=3
## Iteration 506: time=0.01, ELBO=-241395.65, deltaELBO=4.002 (0.00081208%), Factors=3
## Iteration 507: time=0.00, Factors=3
## Iteration 508: time=0.00, Factors=3
## Iteration 509: time=0.00, Factors=3
## Iteration 510: time=0.00, Factors=3
## Iteration 511: time=0.01, ELBO=-241391.74, deltaELBO=3.914 (0.00079430%), Factors=3
## Iteration 512: time=0.00, Factors=3
## Iteration 513: time=0.00, Factors=3
```

```
## Iteration 514: time=0.00, Factors=3
## Iteration 515: time=0.00, Factors=3
## Iteration 516: time=0.01, ELBO=-241387.65, deltaELBO=4.092 (0.00083045%), Factors=3
## Iteration 517: time=0.00, Factors=3
## Iteration 518: time=0.00, Factors=3
## Iteration 519: time=0.00, Factors=3
## Iteration 520: time=0.00, Factors=3
## Iteration 521: time=0.01, ELBO=-241383.35, deltaELBO=4.298 (0.00087219%), Factors=3
## Iteration 522: time=0.01, Factors=3
## Iteration 523: time=0.00, Factors=3
## Iteration 524: time=0.00, Factors=3
## Iteration 525: time=0.00, Factors=3
## Iteration 526: time=0.01, ELBO=-241378.82, deltaELBO=4.535 (0.00092042%), Factors=3
## Iteration 527: time=0.00, Factors=3
## Iteration 528: time=0.00, Factors=3
## Iteration 529: time=0.00, Factors=3
## Iteration 530: time=0.00, Factors=3
## Iteration 531: time=0.01, ELBO=-241374.03, deltaELBO=4.787 (0.00097156%), Factors=3
## Iteration 532: time=0.00, Factors=3
## Iteration 533: time=0.00, Factors=3
## Iteration 534: time=0.00, Factors=3
## Iteration 535: time=0.00, Factors=3
## Iteration 536: time=0.01, ELBO=-241368.61, deltaELBO=5.422 (0.00110041%), Factors=3
## Iteration 537: time=0.00, Factors=3
## Iteration 538: time=0.00, Factors=3
## Iteration 539: time=0.00, Factors=3
## Iteration 540: time=0.00, Factors=3
## Iteration 541: time=0.01, ELBO=-241363.28, deltaELBO=5.323 (0.00108023%), Factors=3
## Iteration 542: time=0.00, Factors=3
## Iteration 543: time=0.00, Factors=3
## Iteration 544: time=0.00, Factors=3
## Iteration 545: time=0.00, Factors=3
## Iteration 546: time=0.01, ELBO=-241359.80, deltaELBO=3.486 (0.00070749%), Factors=3
## Iteration 547: time=0.00, Factors=3
## Iteration 548: time=0.00, Factors=3
## Iteration 549: time=0.00, Factors=3
## Iteration 550: time=0.00, Factors=3
## Iteration 551: time=0.01, ELBO=-241355.25, deltaELBO=4.548 (0.00092301%), Factors=3
## Iteration 552: time=0.00, Factors=3
## Iteration 553: time=0.00, Factors=3
## Iteration 554: time=0.00, Factors=3
## Iteration 555: time=0.00, Factors=3
## Iteration 556: time=0.01, ELBO=-241350.76, deltaELBO=4.485 (0.00091017%), Factors=3
## Iteration 557: time=0.00, Factors=3
## Iteration 558: time=0.00, Factors=3
## Iteration 559: time=0.00, Factors=3
## Iteration 560: time=0.00, Factors=3
## Iteration 561: time=0.01, ELBO=-241346.70, deltaELBO=4.065 (0.00082487%), Factors=3
## Iteration 562: time=0.00, Factors=3
## Iteration 563: time=0.00, Factors=3
## Iteration 564: time=0.00, Factors=3
## Iteration 565: time=0.00, Factors=3
## Iteration 566: time=0.01, ELBO=-241344.11, deltaELBO=2.588 (0.00052527%), Factors=3
## Iteration 567: time=0.00, Factors=3
```

```
## Iteration 568: time=0.00, Factors=3
## Iteration 569: time=0.00, Factors=3
## Iteration 570: time=0.00, Factors=3
## Iteration 571: time=0.01, ELBO=-241343.04, deltaELBO=1.070 (0.00021718%), Factors=3
## Iteration 572: time=0.00, Factors=3
## Iteration 573: time=0.01, Factors=3
## Iteration 574: time=0.01, Factors=3
## Iteration 575: time=0.00, Factors=3
## Iteration 576: time=0.01, ELBO=-241341.74, deltaELBO=1.297 (0.00026316%), Factors=3
##
## Converged!
##
##
##
## ########################
## ## Training finished ##
## ########################
##
##
## Warning: Output file /home/adam/Desktop/China_Workshop/training_multiomics/mofa/model.hdf5 already ex
## Saving model in /home/adam/Desktop/China_Workshop/training_multiomics/mofa/model.hdf5...

## Warning in .quality_control(object, verbose = verbose): Factor(s) 1, 3 are strongly correlated with t
```

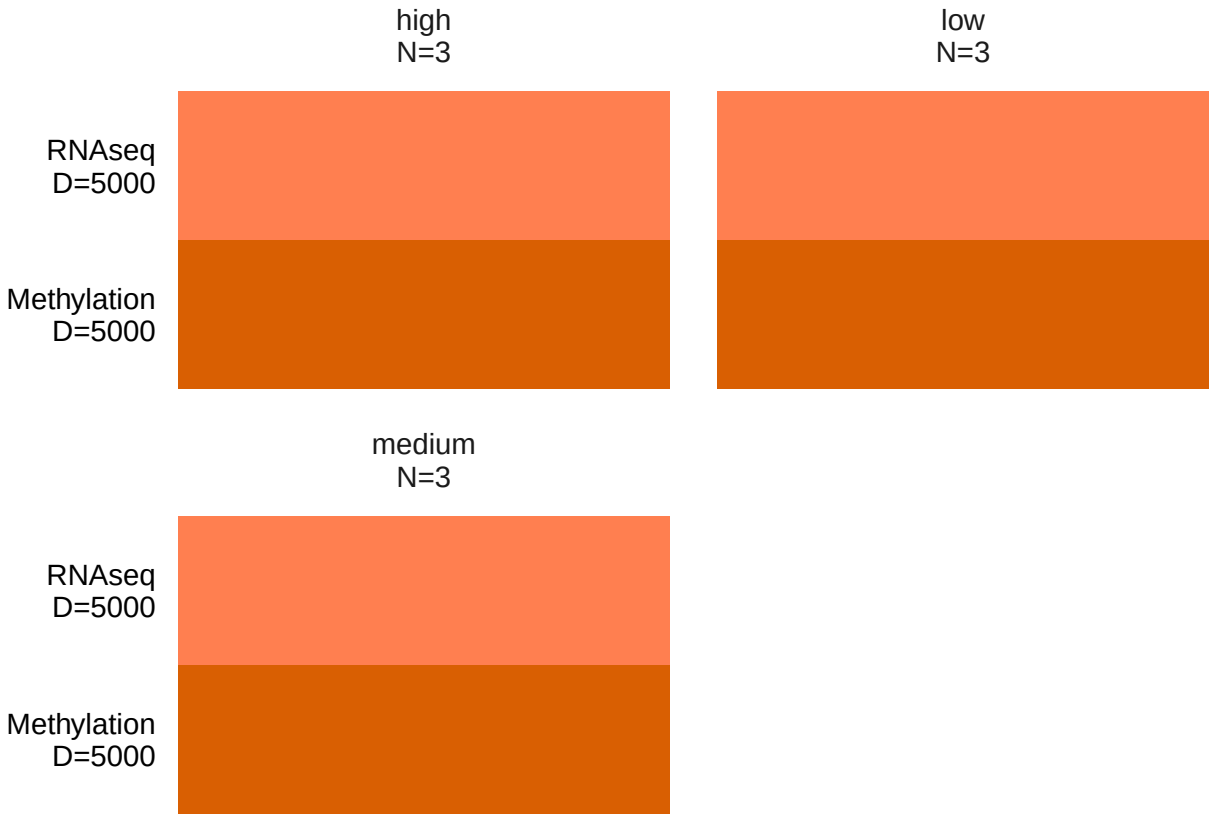
---

8. Inspect variance explained by the trained model

The first step in the MOFA analysis is to quantify the amount of variance explained (R2) by each factor in each data modality. The variance explained estimates are stored in the hdf5 file and loaded in `mofa_trained@cache$variance_explained$r2_per_factor[[1]]`. As observed, RNA-seq liver data has an important cumulative variance explained of 52 % and methylation data of 32.43 % among the three observed factors. We will inspect in the future cells the biomarkers driven by the Views (Modalities) across groups differentiation.

```
# ------------------------------------
# Downstream Analysis & Visualization
# ------------------------------------

### Check the Number of features according to groups
plot_data_overview(mofa_trained)
```



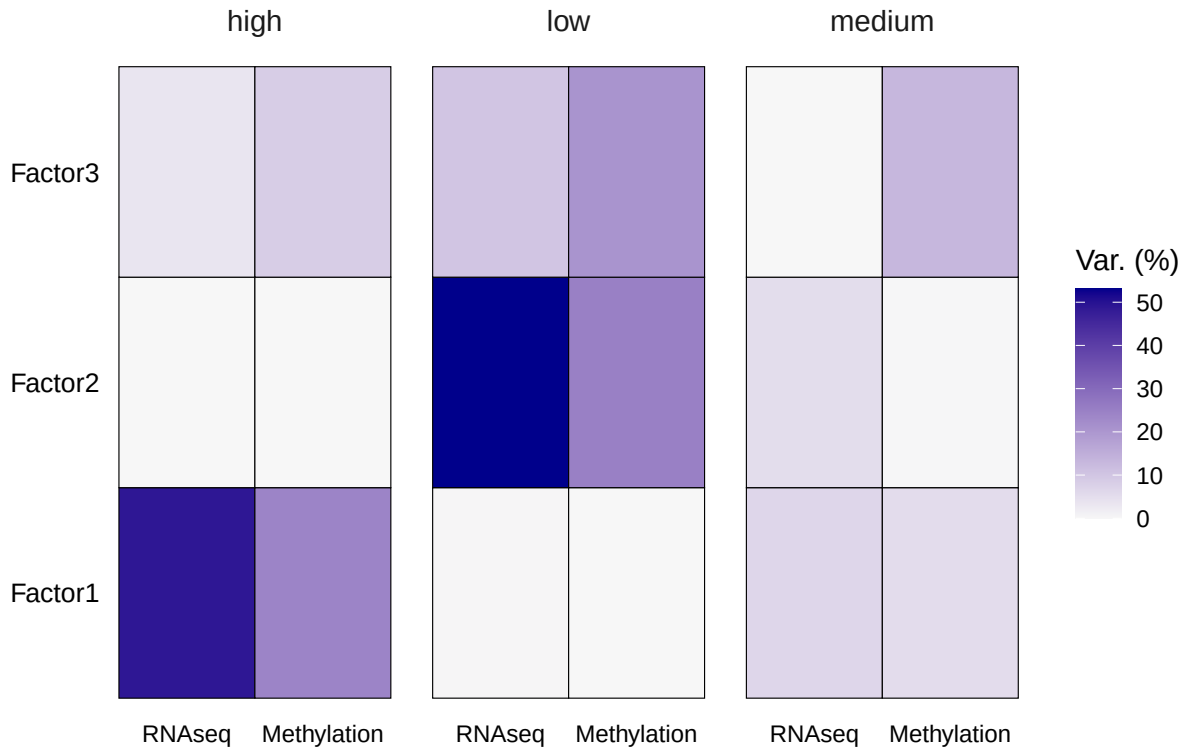

```
# Total variance explained per Omics
print(mofa_trained@cache$variance_explained$r2_total[[1]])
```

```
##      RNAseq Methylation
##    52.08733    32.43441
```

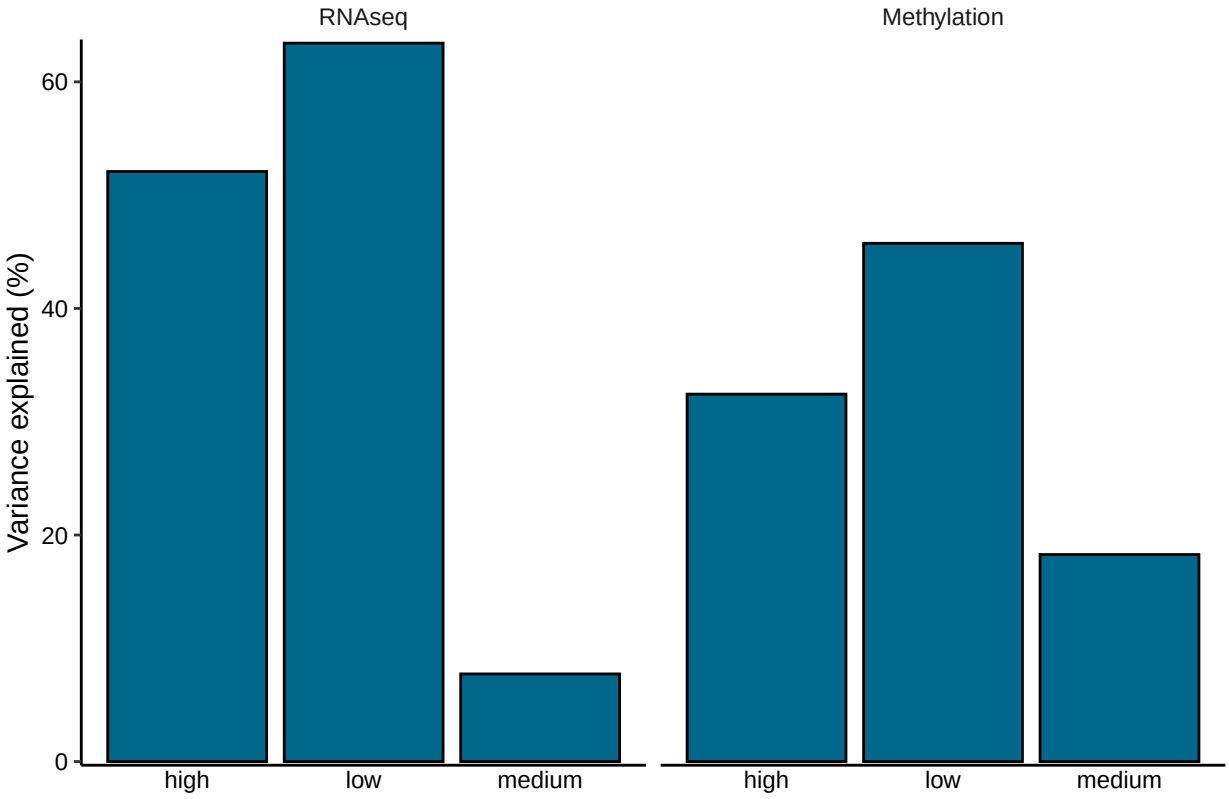
```
# Variance explained by each factor per view
head(mofa_trained@cache$variance_explained$r2_per_factor[[1]])
```

```
##               RNAseq   Methylation
## Factor1 4.889330e+01  2.429984e+01
## Factor2 2.384186e-05 -3.576279e-05
## Factor3 3.685707e+00  8.536720e+00
```

```
###Plot variance
plot_variance_explained(mofa_trained, x="view", y="factor") # Modality
```




```
plot_variance_explained(mofa_trained, x="group", y="factor", plot_total = T)[[2]] # Group
```



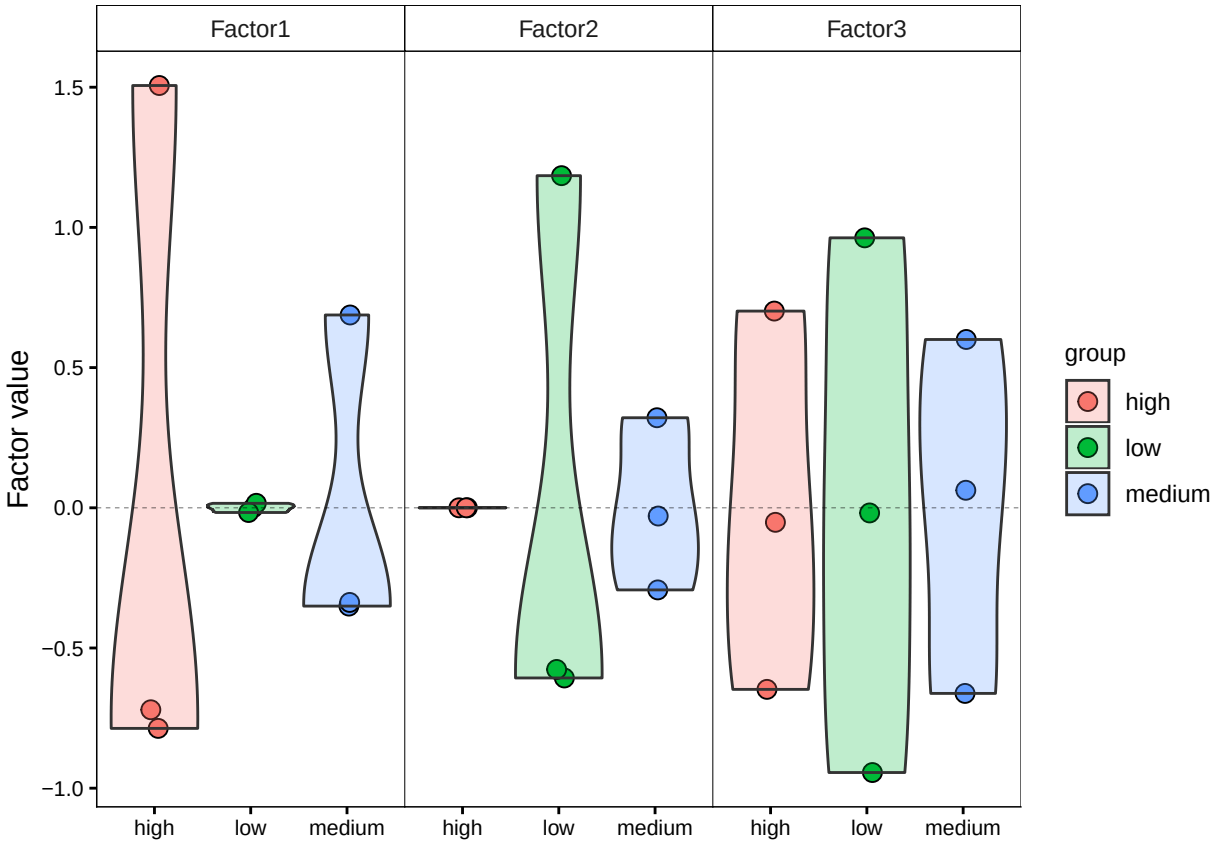
9. Plot the MOFA factors by group

MOFA factors are assigned to each sample observed by MOFA trained model. It describes for each sample the molecular variation pattern observed among the integrated data. The values are centred around zero; therefore, their absolute numerical values are less important than the relative positions of the samples. Samples with factor values close to `0.0` indicates that there is no consistent distinction among the samples observed in the respective latent factors.

In this chunk of code, `factors = c(1,2)` produces separate panels for Factor 1, Factor 2 and Factor 3. The argument `color_by = "group"` colours the samples according to their predefined High, Low, and Medium boar-taint groups. This colouring is used for downstream interpretation and `does not mean that the groups were used as supervised outcomes when the factors were learned`.

For Factor 1, the Low taint-group samples are concentrated around zero, whereas the High taint-group samples display greater dispersion. The Medium group includes both negative and positive factor values. For Factor 2, the High taint-group samples are mainly concentrated near zero, while the Low group contains negative factor values together with one strongly positive value. The Medium group spans from moderately negative to positive factor values. For Factor 3, heterogeneous variability is observed among individuals within each of the three taint groups, with the Low-taint group showing the widest distribution of factor values. As indicated by the previous plot, methylation features contribute more strongly to this variability than RNA-seq features, suggesting the presence of substantial inter-individual methylation differences within and across the taint groups. The differentiation between the RNA-seq and methylation views from the different observed groups highlighted in the previous analysis is also reflected in the distinct factor values assigned to the individual samples.

```
plot_factor(mofa_trained,
                  factors = 1:3,
                  color_by = "group",
                  dot_size = 3,         # change dot size
                  dodge = T,            # dodge points with different colors
                  legend = T,
                  add_violin = T,
                  violin_alpha = 0.25)
```



10. Plot Factor 1 against Factor 2 - Factor 1 against Factor 3

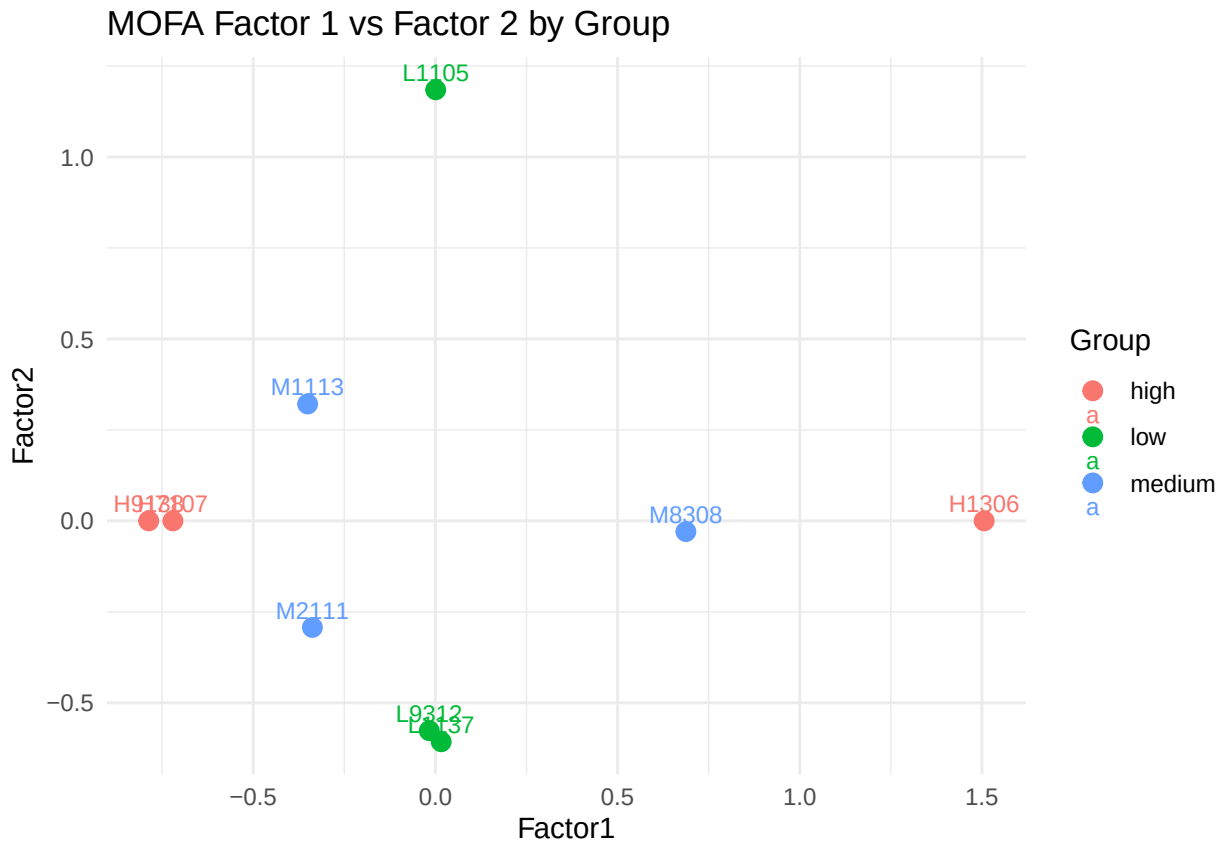
Comparison of the factor scatter plots with the previous group-wise factor distributions confirms inter-individual variability within the three taint groups. Factor 1 primarily distinguishes samples from the High- and Medium-taint groups from the more tightly clustered Low-taint samples, with `H1306` showing the most distinct Factor 1 score among the High-taint individuals. Along Factor 2, the Low-taint samples display greater dispersion and therefore contribute prominently to the observed sample separation. The Factor 1 versus Factor 3 plot reveals a different structure. Although `H9138` and `H7107` were less differentiated along Factor 1, they occupy more distinct positions along Factor 3. This indicates that Factor 3 captures a source of molecular variability that differs from the variation represented by Factor 1. Based on the variance explained by each factor within each omics view, this Factor 3 pattern may be more strongly associated with methylation variability than with RNA-seq variability.

```
# factors scatter plot
# Convert long to wide format
```

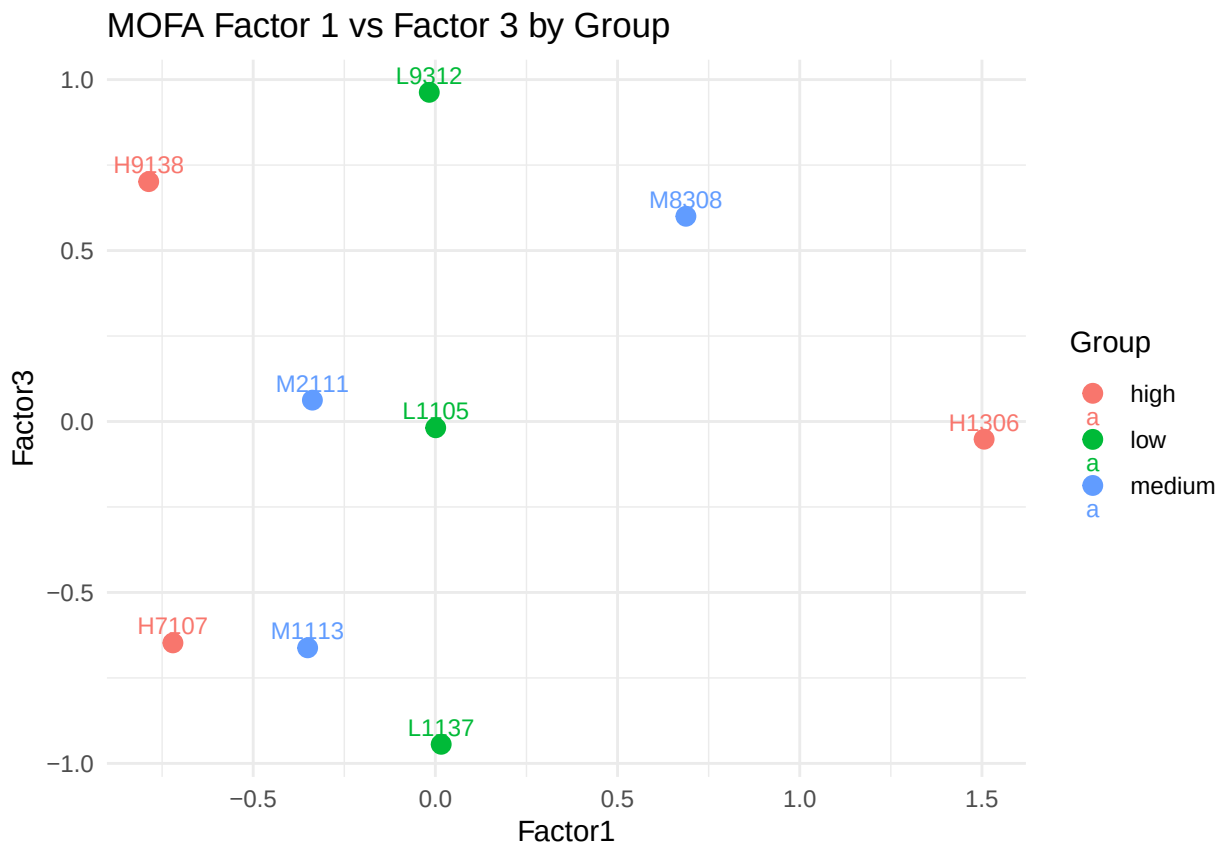
```r
factors_df <- get_factors(mofa_trained, factors = "all", as.data.frame = TRUE)

factors_wide <- factors_df %>%
  pivot_wider(names_from = factor, values_from = value)

# Scatter plot F1 and F2
ggplot(factors_wide, aes(x = Factor1, y = Factor2, color = group)) +
  geom_point(size = 3) +
  geom_text(aes(label = sample), vjust = -0.5, hjust = 0.5, size = 3) +  # sample labels above points
  theme_minimal() +
  labs(title = "MOFA Factor 1 vs Factor 2 by Group", color = "Group")
```




```r
# Scatter plot F1 and F3
ggplot(factors_wide, aes(x = Factor1, y = Factor3, color = group)) +
  geom_point(size = 3) +
  geom_text(aes(label = sample), vjust = -0.5, hjust = 0.5, size = 3) +  # sample labels above points
  theme_minimal() +
  labs(title = "MOFA Factor 1 vs Factor 3 by Group", color = "Group")
```



11. Inspect RNA-seq feature weights for Factor 1

In the following plot, Here we investigate the contribution of top features weights in RNA-seq omic modality in Factor 1. From a transcriptomic perspective, the top 5000 biomarkers were restricted to the top 10 features that explains the major differentiations observed across the groups :

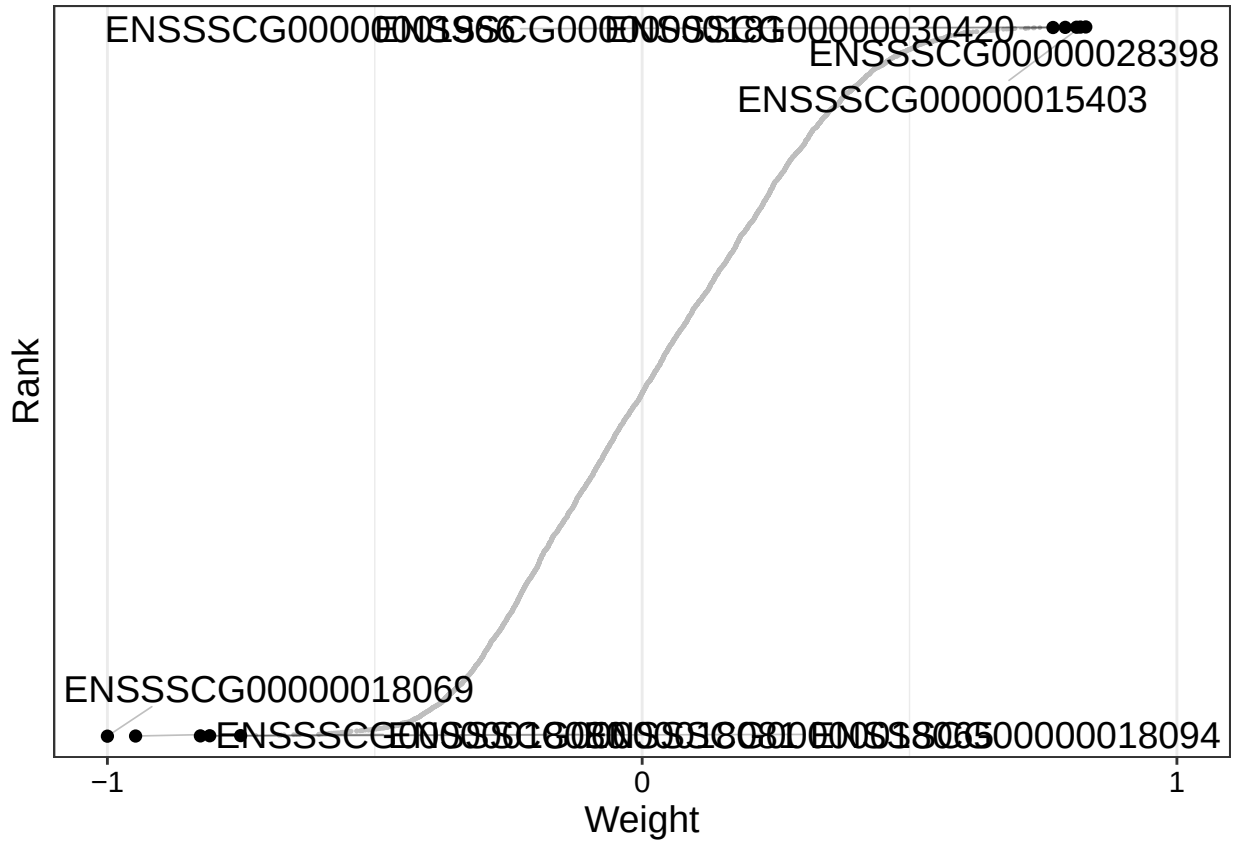
- A positive weight means that the expression of gene (current observed biomarker) tends to be higher in samples with positive values of that factor and lower in samples with negative factor values.
- A negative weight means that its expression tends to be higher in samples with negative values of that factor and lower in samples with positive factor values.
- A weight close to zero indicates that the feature is weakly related to that factor.
- A larger absolute weight indicates a stronger relationship with the factor.

Respectively to the same interpretation approach, methylation data can be seen in the same method. For the top feature in Factor1, the accessibility of the methylation region `chr6_13663616` has a negative weight, indicating that its methylation is higher in samples observed on factors scores that have higher negative scores, compared to samples with positive factors. The observed weight of the feature is currently the highest, indicating a major contribution from the methylation modality in explaining intra-individual differentiation from the current observed biomarker. To investigate other factors in intra-group/intra-individual differentiation, the flag `factor = 1` and `features = 10` can be edited for higher number of variable measurements and other latent factors observed.

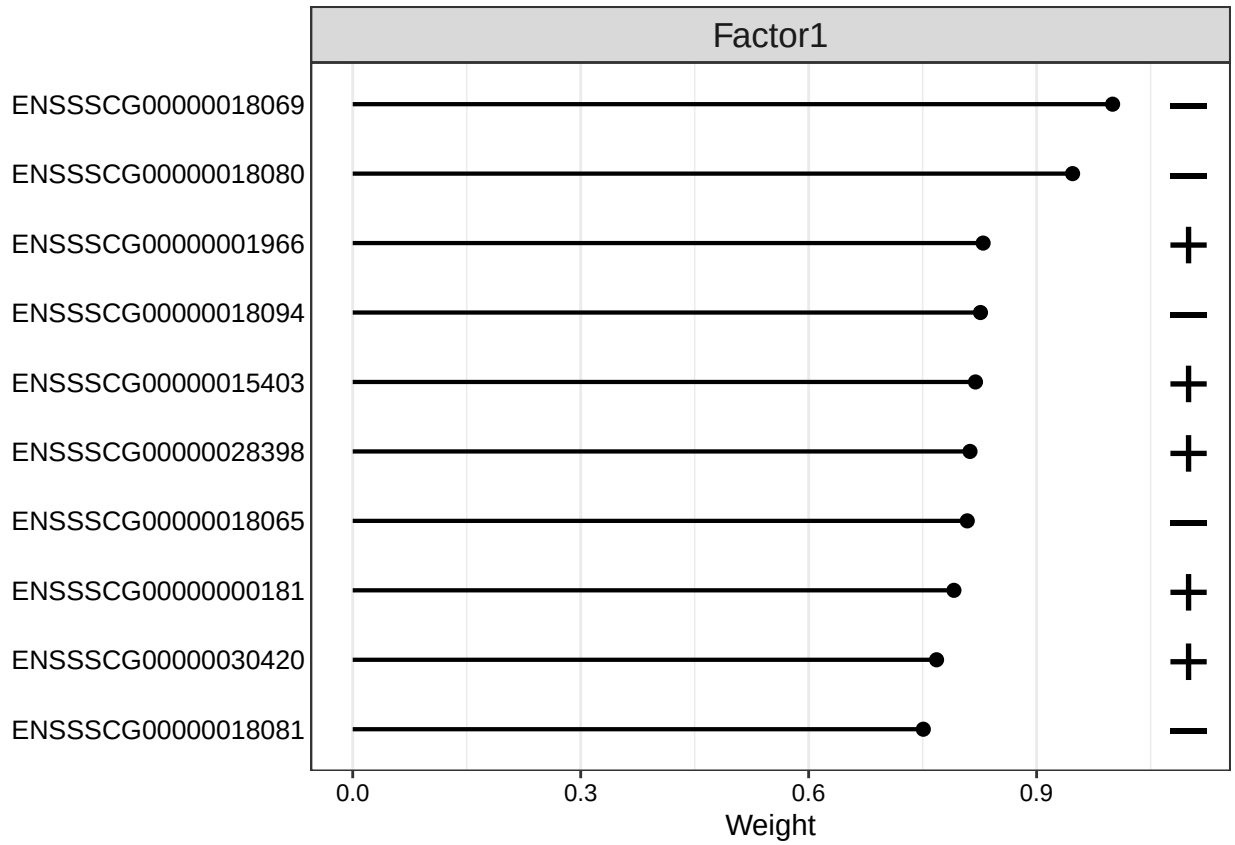
```
################

# Identify Important Genes/CpGs for Each Factor
```

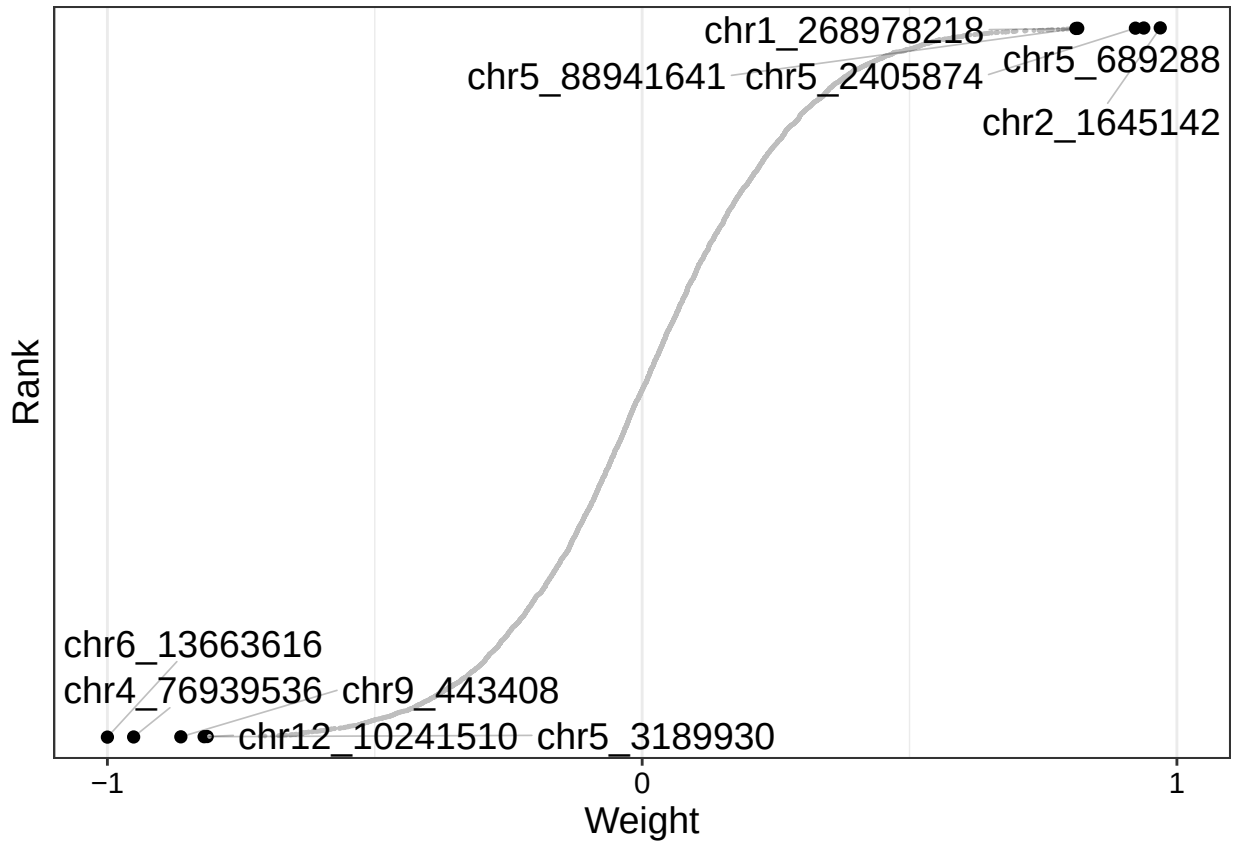
```
plot_weights(mofa_trained,
             view = "RNAseq",
             factor = 1,
             nfeatures = 10,      # Number of features to highlight
             scale = T,
             abs = F              # Take the absolute value?
)
```



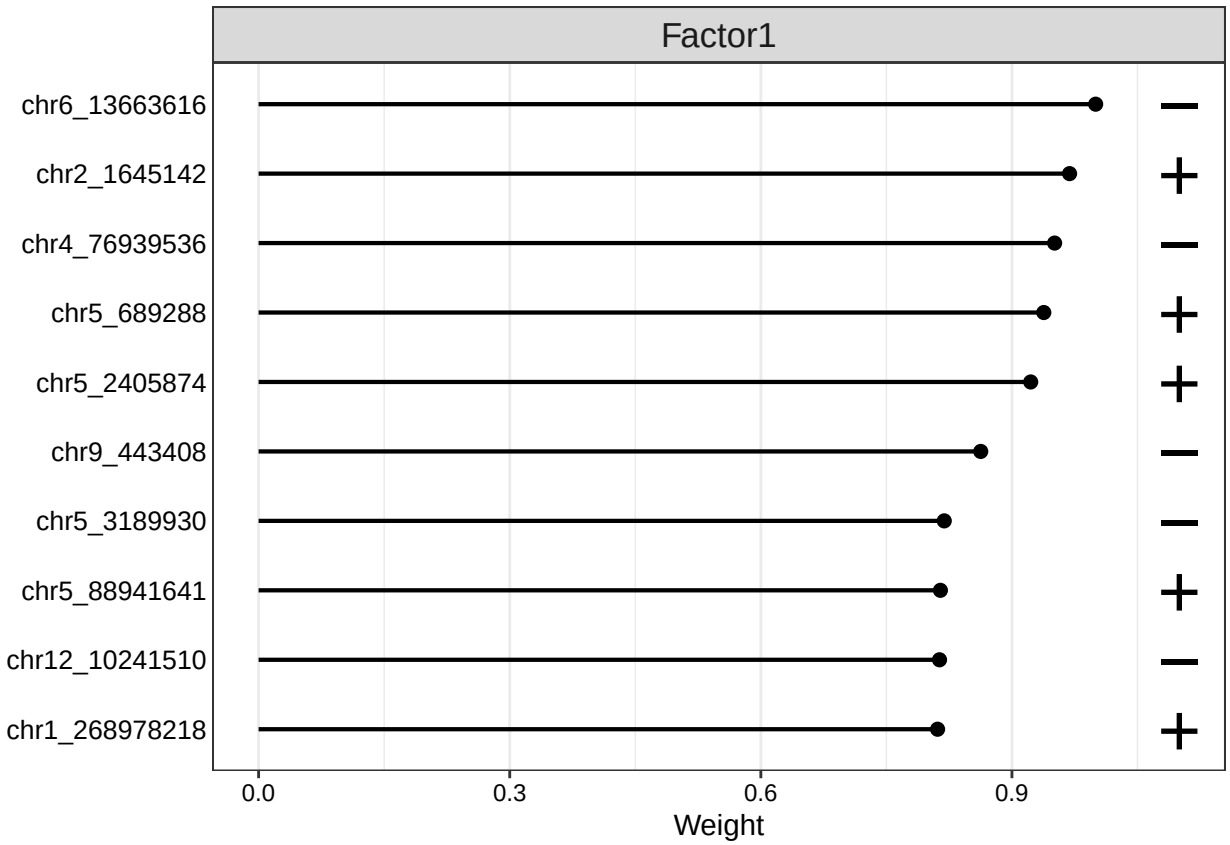

```
# Plot top absolute weights (most influential features regardless of direction)
plot_top_weights(mofa_trained,
                 view = "RNAseq",
                 factor = 1,
                 nfeatures = 10
)
```




```
#### Methylation
# Plot feature weights for Factor 1 in the Methylation view
plot_weights(mofa_trained,
             view = "Methylation",
             factor = 1,
             nfeatures = 10,     # Number of features to highlight
             scale = T,          # Scale weights from -1 to 1
             abs = F             # Take the absolute value?
)
```




```
# Plot top absolute weights (most influential features regardless of direction)
plot_top_weights(mofa_trained,
                 view = "Methylation",
                 factor = 1,
                 nfeatures = 10
)
```



12. Relate the top RNA-seq feature to Factor 1-2-3 (Scatter plots)

This analysis examines the relationship between the sample scores for Factor 1 and the expression values of the RNA-seq feature with the largest absolute weight on that factor. MOFA2 recommends plotting current observed feature (in this example plots, we take the highest weighted variable) values against factor scores to visualize the coordinated molecular variability captured by the model. A linear regression line can be added to assess whether the relationship between the selected feature and the latent factor is approximately linear.

In this scatter plot, the x-axis represents the Factor 1 score and the y-axis represents the expression level of the selected RNA-seq feature. Samples are coloured according to their taint group. The fitted linear regression for each group is observed by the correlation between the feature selected (Top gene selected) to the Factor 1 score.

On the following plot, Latent factors capture major sources of variation across the different data modalities, and each individual is assigned a factor score that determines its position along the corresponding latent factor. Because the selected gene has a negative weight, its expression tends to be higher in samples with negative factor scores and lower in samples with positive factor scores, within this fitted model. To confirm these results, we have plotted another heatmap that investigates covariation patterns in the input matrices. In fact, by observing the gene `ENSSSCG00000018069` in the heatmap, it is quite observable the way how the gene expression distinguishes in between the 3 high-boar taints phenotype is.

```
#######  Gene correlates with factor 1 component ################################

# Extract weights
```

```r
weights <- get_weights(mofa_trained, views = "RNAseq", factors = 1, as.data.frame = TRUE)
#These weights represent how strongly each feature contributes to Factor 1

top_genes <- get_weights(mofa_trained,
                         view = "RNAseq",
                         factor = 1,
                         as.data.frame = TRUE)

top_feature <- top_genes[order(abs(top_genes$value), decreasing = TRUE), ]
top_feature <- weights[order(-abs(weights$value)), "feature"][1]
head(top_feature)
```

```
## [1] ENSSSCG00000018069
## 10000 Levels: ENSSSCG00000018700 ENSSSCG00000020505 ENSSSCG00000023855 ENSSSCG00000022808 ENSSSCG0000
```

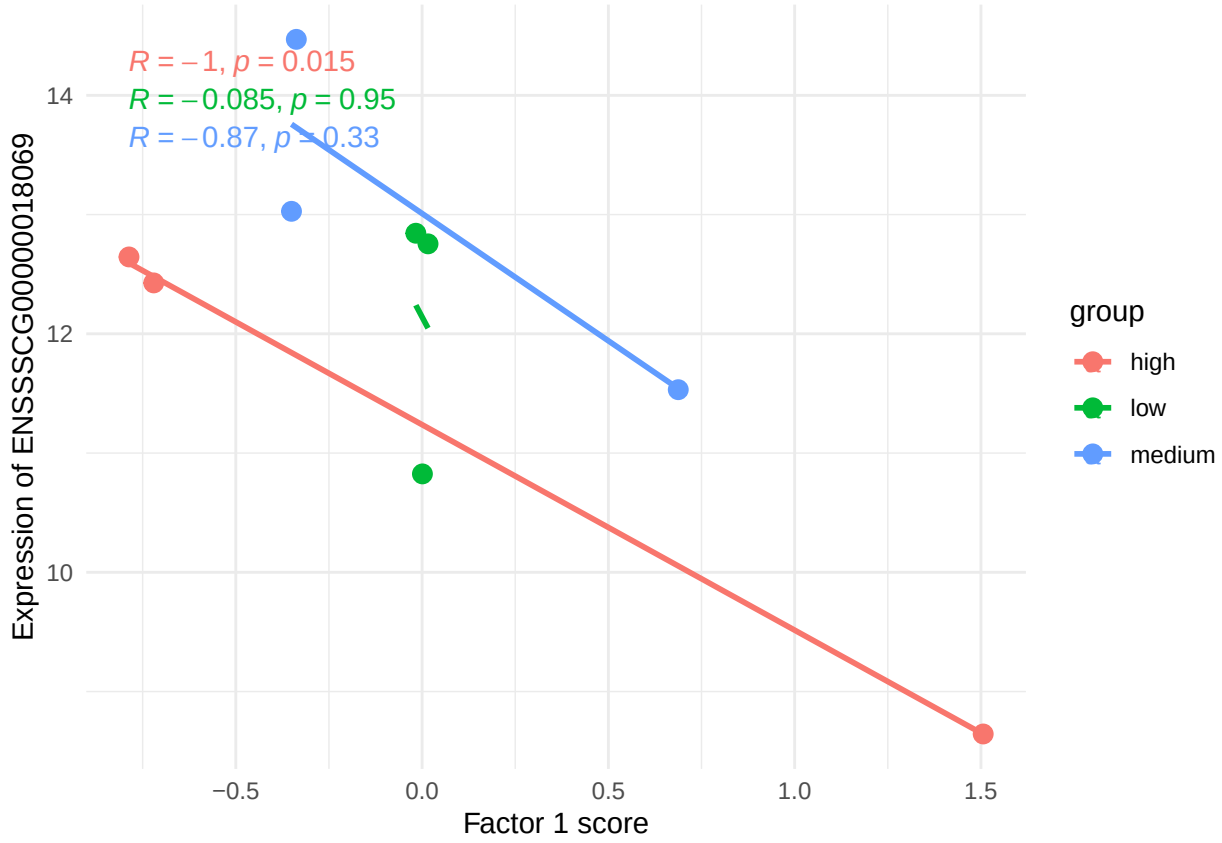
```r
# Extract factor scores
factors <- get_factors(mofa_trained, factors = 1, as.data.frame = TRUE)

# Extract feature expression
data <- get_data(mofa_trained, view = "RNAseq", as.data.frame = TRUE)
feature_data <- subset(data, as.character(feature) == as.character(top_feature))

# Merge data
df <- merge(factors, feature_data, by = "sample")
df$group <- samples_metadata(mofa_trained)$group

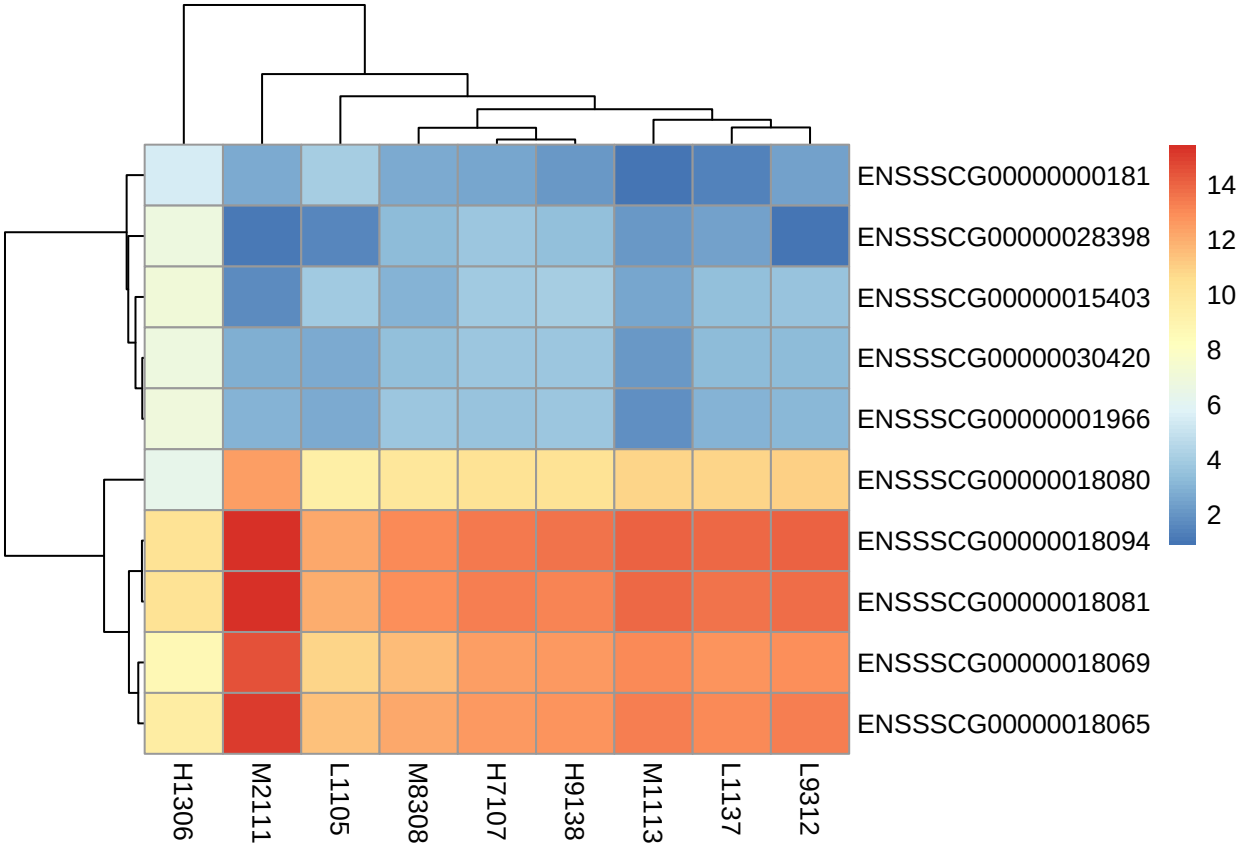
# Plot
ggplot(df, aes(x = value.x, y = value.y, color = group)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  stat_cor(method = "pearson") +
  labs(
    x = "Factor 1 score",
    y = paste("Expression of", top_feature)
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
plot_data_heatmap(mofa_trained,
  view = "RNAseq",         # view of interest
  factor = 1,              # factor of interest
  features = 10,           # number of features to plot (they are selected by weight)

  # extra arguments that are passed to the `pheatmap` function
  cluster_rows = TRUE, cluster_cols = TRUE,
  show_rownames = TRUE, show_colnames = TRUE
)
```




```
## Obseving the weight of the expressiong gene, the sign of it is negative. Which means: the expression
## for samples that have negative factore score (value), and the expression of this gene seems to be lo
## positive factor score.
```

```
#######  Gene correlates with factor 2 component ################################

# Extract weights
weights <- get_weights(mofa_trained, views = "RNAseq", factors = 2, as.data.frame = TRUE)
#These weights represent how strongly each feature contributes to Factor 1

top_genes <- get_weights(mofa_trained,
                         view = "RNAseq",
                         factor = 2,
                         as.data.frame = TRUE)

top_feature <- top_genes[order(abs(top_genes$value), decreasing = TRUE), ]
top_feature <- weights[order(-abs(weights$value)), "feature"][1]
head(top_feature)
```

```
## [1] ENSSSCG00000023880
## 10000 Levels: ENSSSCG00000018700 ENSSSCG00000020505 ENSSSCG00000023855 ENSSSCG00000022808 ENSSSCG0000
```

```
# Extract factor scores
factors <- get_factors(mofa_trained, factors = 2, as.data.frame = TRUE)

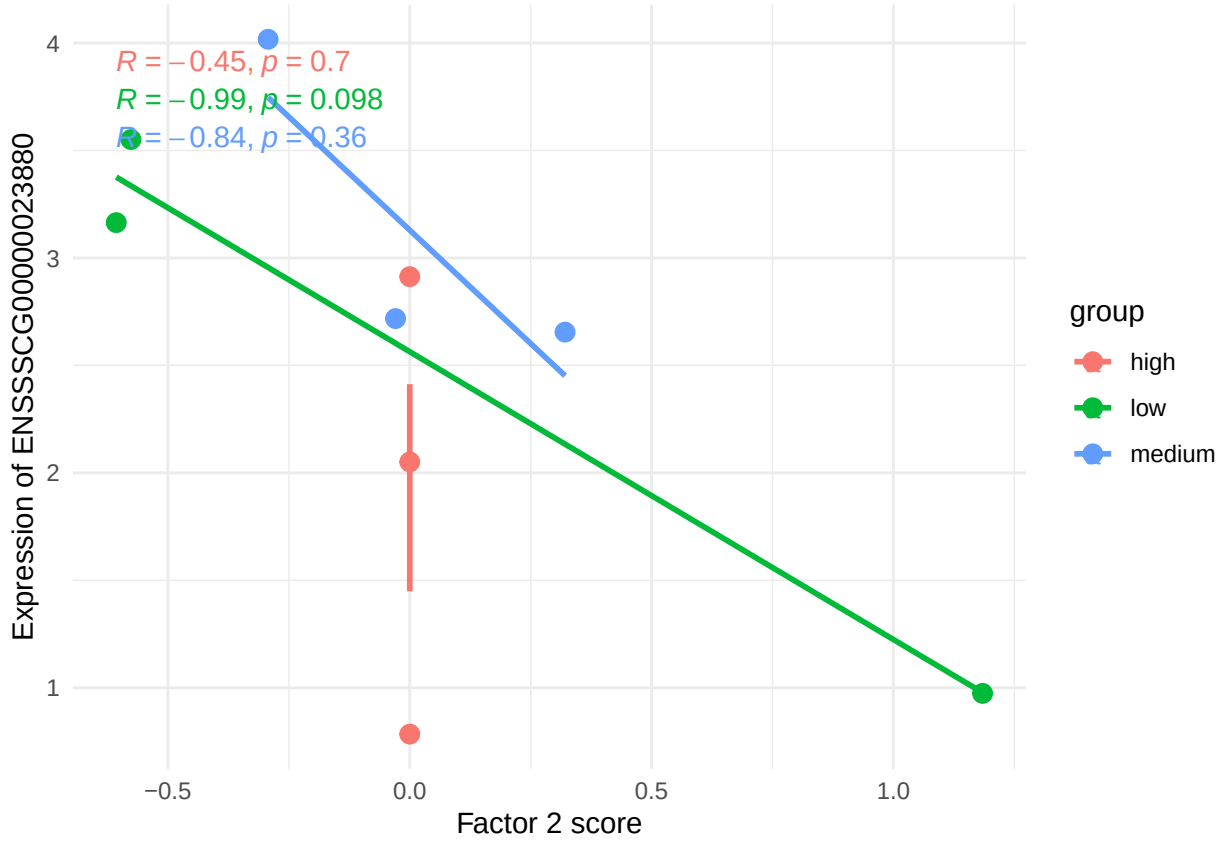
# Extract feature expression
data <- get_data(mofa_trained, view = "RNAseq", as.data.frame = TRUE)
```

```r
feature_data <- subset(data, as.character(feature) == as.character(top_feature))

# Merge data
df <- merge(factors, feature_data, by = "sample")
df$group <- samples_metadata(mofa_trained)$group

# Plot
ggplot(df, aes(x = value.x, y = value.y, color = group)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  stat_cor(method = "pearson") +
  labs(
    x = "Factor 2 score",
    y = paste("Expression of", top_feature)
  ) +
  theme_minimal()
```

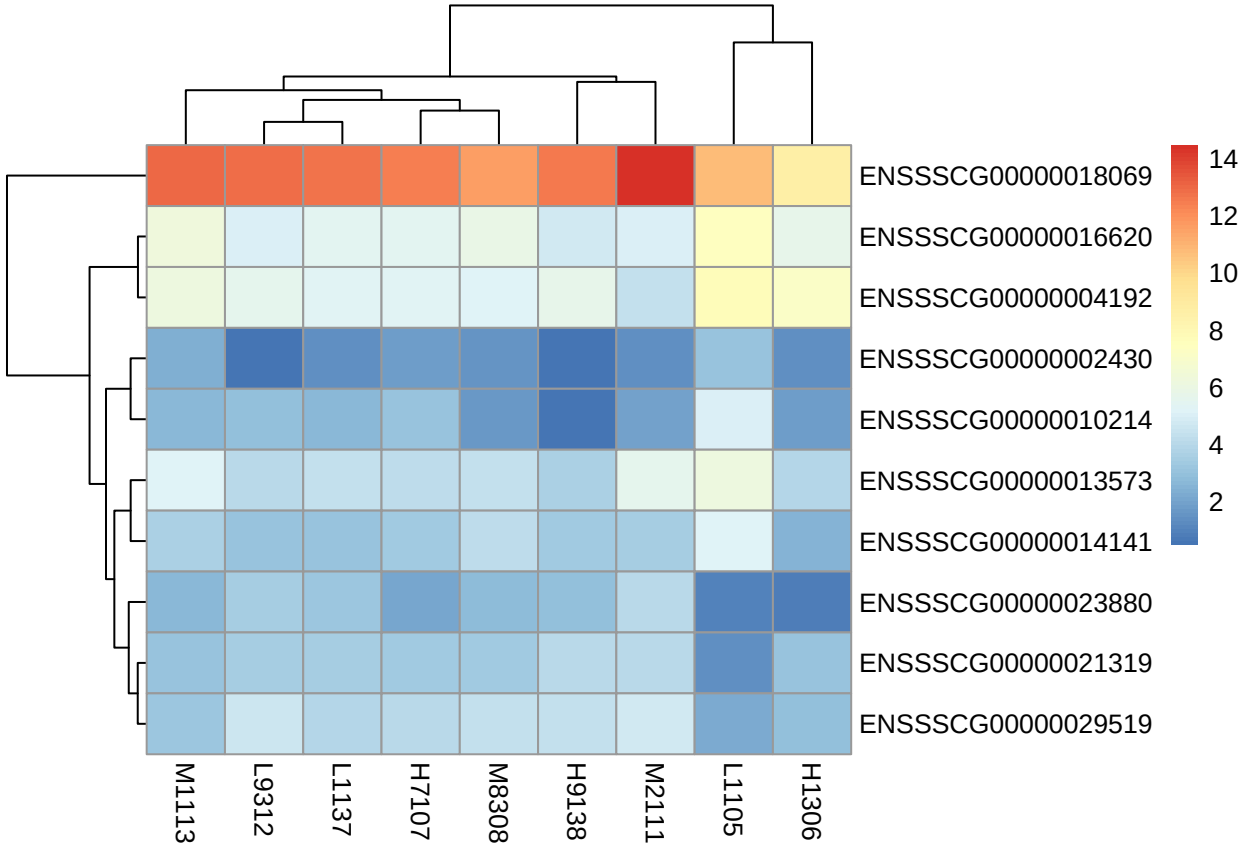
```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
plot_data_heatmap(mofa_trained,
  view = "RNAseq",          # view of interest
  factor = 2,               # factor of interest
  features = 10,            # number of features to plot (they are selected by weight)

  # extra arguments that are passed to the `pheatmap` function
  cluster_rows = TRUE, cluster_cols = TRUE,
```

```
  show_rownames = TRUE, show_colnames = TRUE
)
```




```
#######  Gene correlates with factor 3 component ################################

# Extract weights
weights <- get_weights(mofa_trained, views = "RNAseq", factors = 3, as.data.frame = TRUE)
#These weights represent how strongly each feature contributes to Factor 1

top_genes <- get_weights(mofa_trained,
                         view = "RNAseq",
                         factor = 3,
                         as.data.frame = TRUE)

top_feature <- top_genes[order(abs(top_genes$value), decreasing = TRUE), ]
top_feature <- weights[order(-abs(weights$value)), "feature"][1]
head(top_feature)
```

```
## [1] ENSSSCG00000011482
## 10000 Levels: ENSSSCG00000018700 ENSSSCG00000020505 ENSSSCG00000023855 ENSSSCG00000022808 ENSSSCG0000
```

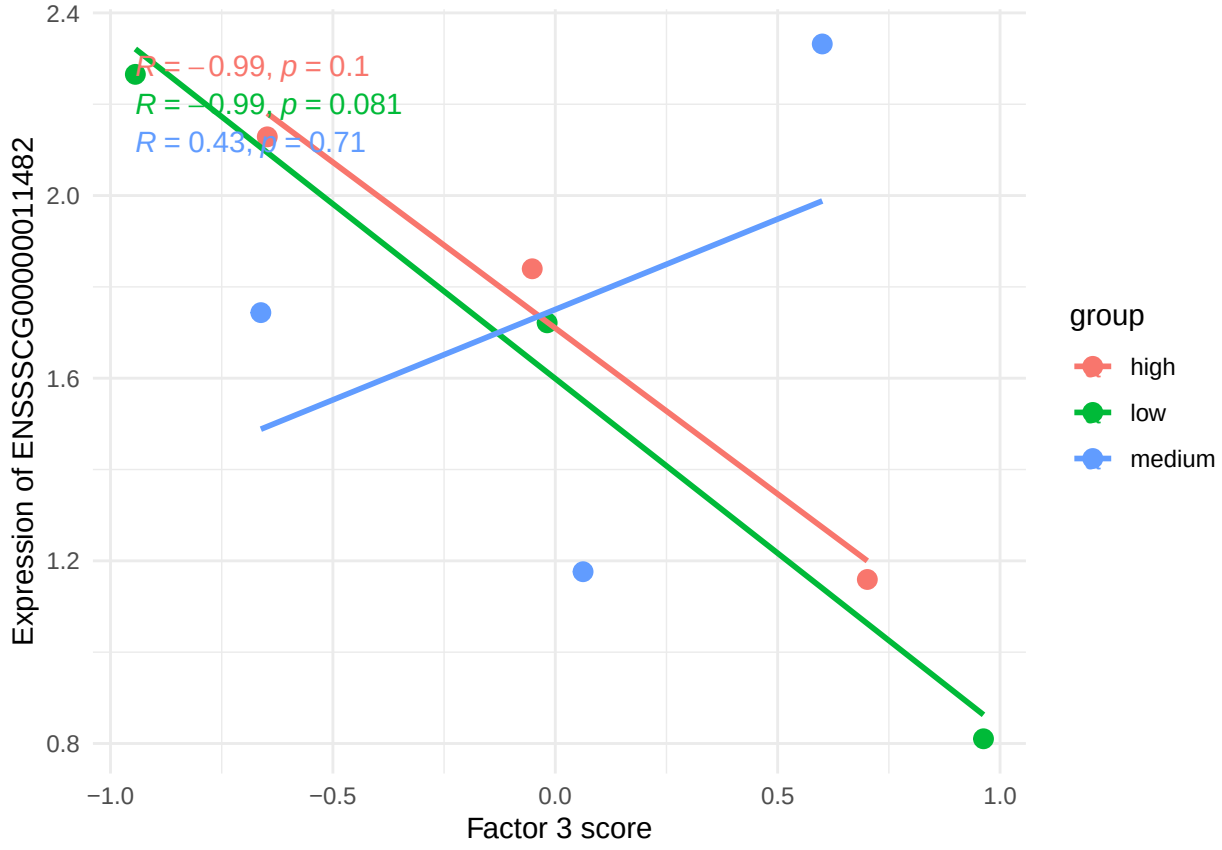
```
# Extract factor scores
factors <- get_factors(mofa_trained, factors = 3, as.data.frame = TRUE)

# Extract feature expression
data <- get_data(mofa_trained, view = "RNAseq", as.data.frame = TRUE)
feature_data <- subset(data, as.character(feature) == as.character(top_feature))
```

```r
# Merge data
df <- merge(factors, feature_data, by = "sample")
df$group <- samples_metadata(mofa_trained)$group

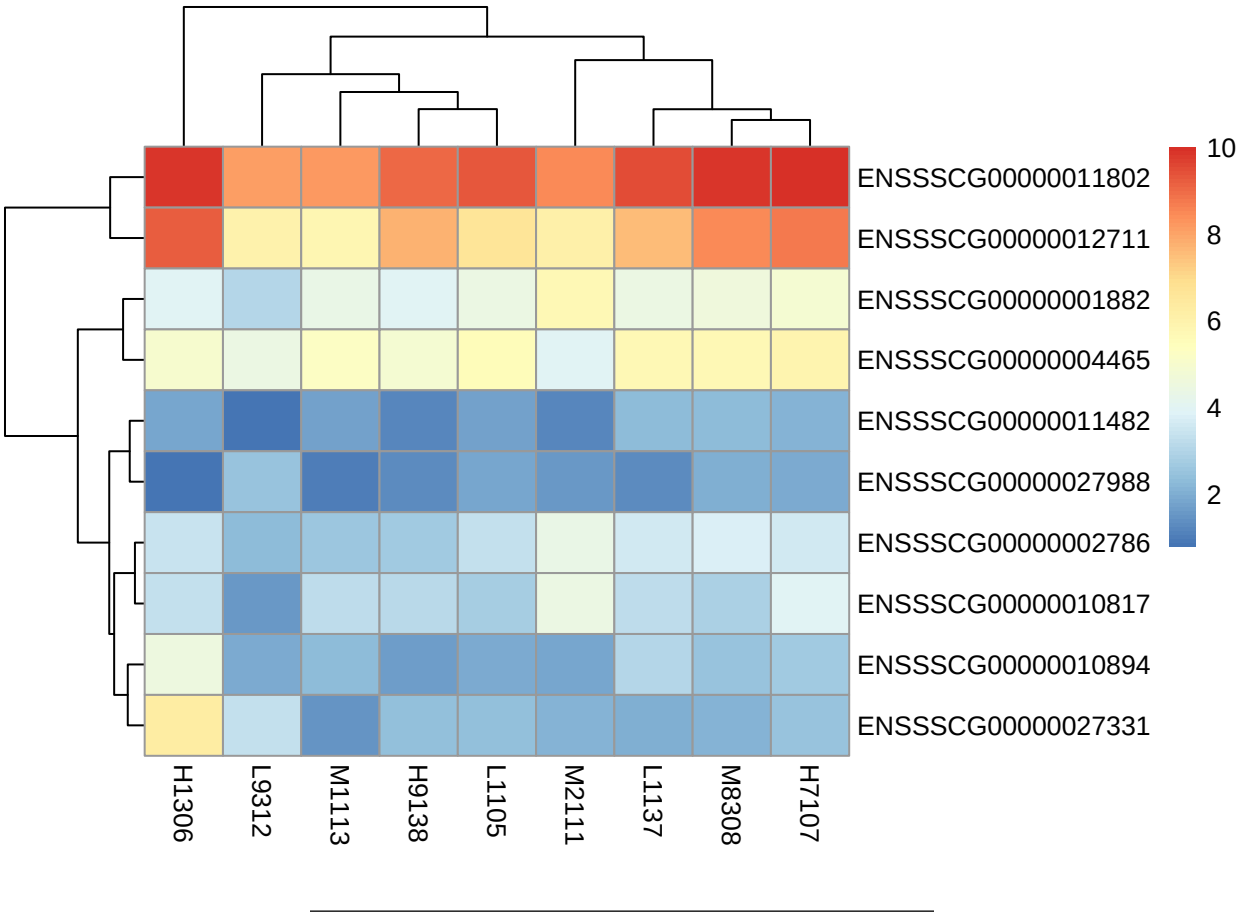
# Plot
ggplot(df, aes(x = value.x, y = value.y, color = group)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  stat_cor(method = "pearson") +
  labs(
    x = "Factor 3 score",
    y = paste("Expression of", top_feature)
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
plot_data_heatmap(mofa_trained,
  view = "RNAseq",         # view of interest
  factor = 3,              # factor of interest
  features = 10,           # number of features to plot (they are selected by weight)

  # extra arguments that are passed to the `pheatmap` function
  cluster_rows = TRUE, cluster_cols = TRUE,
  show_rownames = TRUE, show_colnames = TRUE
)
```



13. Visualize the top features driving Factor 1

The violin plots display the observed values of the RNA-seq and methylation features with the largest absolute weights for Factor 1. These features are therefore the variables most strongly associated with the covariation pattern represented by this factor within their respective omics views. For the RNA-seq view, the plots show the normalized expression values of the selected genes across the three boar-taint groups. For the methylation view, they show methylation M-values. Some features display relatively similar values across the cohort, whereas others show pronounced differences among individuals or groups (All the group commonly different to the rest of the measured phenotypes categories, or individuals are different within the group). These patterns indicate inter-individual molecular variability associated with Factor 1.

As a multivariate latent-factor approach, MOFA represents each omics modality as a linear combination of latent factors and feature-specific weights. This is conceptually related to PCA, which summarizes variation through linear principal components, but MOFA extends this framework to multiple data modalities. In fact, what we have presented in this tutorial is just a linear modelization of a unsupervised approach. MOFA can go beyond this by modelling also non-linear relationships. I would invite to read these 2 links about MOFA: `https://pmc.ncbi.nlm.nih.gov/articles/PMC6010767/` and `https://biofam.github.io/MOFA2/`.

```
# Visualization of Top 30 Features Driving Latent Factor 1 in MOFA (RNA-Seq and Methylation)
# Get latent factors (samples × factors)
factors <- get_factors(mofa_trained, factors = "all")

# Get feature weights for all views (features × factors)
weights <- get_weights(mofa_trained, views = c("RNAseq", "Methylation"), factors = 1)
```

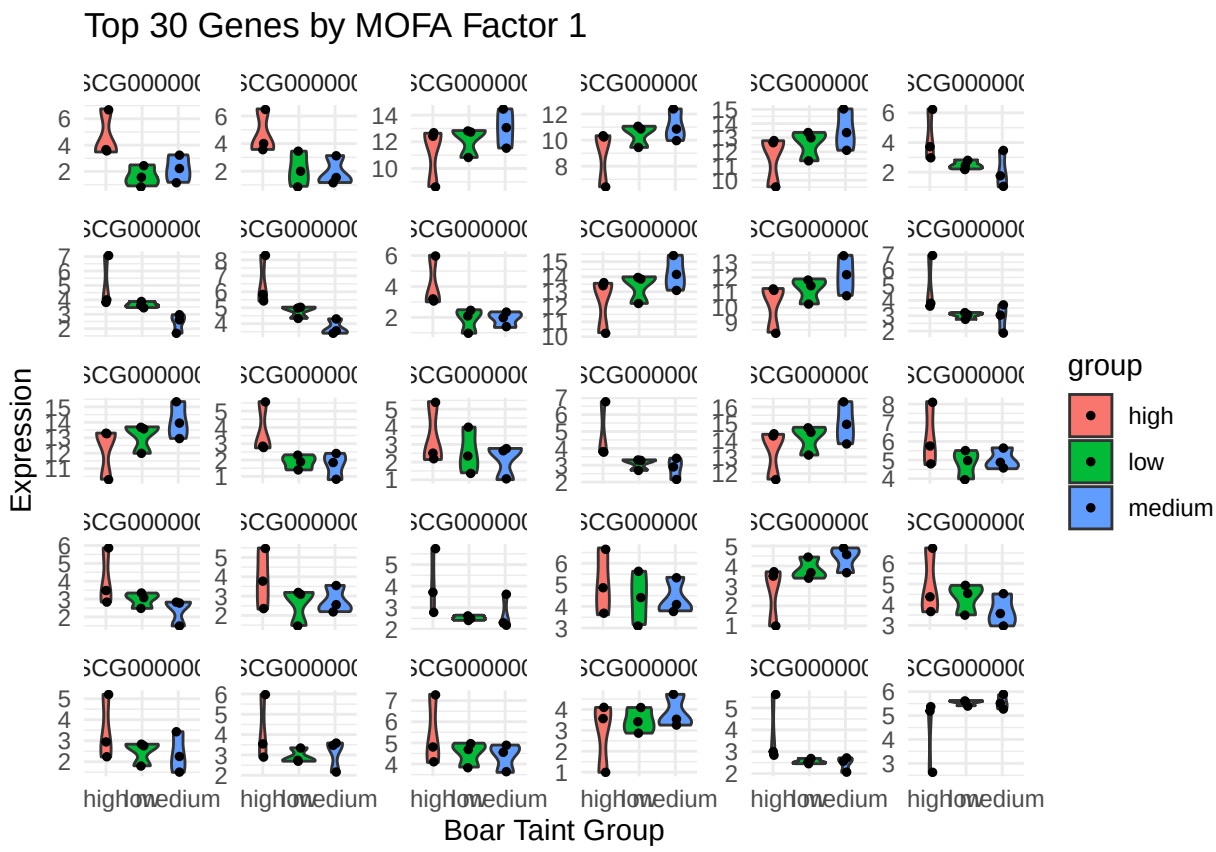
```r
# Top 30 genes with highest absolute weights in RNAseq
top_genes <- sort(abs(weights$RNAseq[,1]), decreasing = TRUE)[1:30]
top_genes_names <- names(top_genes)

# Top 30 CpGs with highest absolute weights in Methylation
top_cpgs <- sort(abs(weights$Methylation[,1]), decreasing = TRUE)[1:30]
top_cpgs_names <- names(top_cpgs)

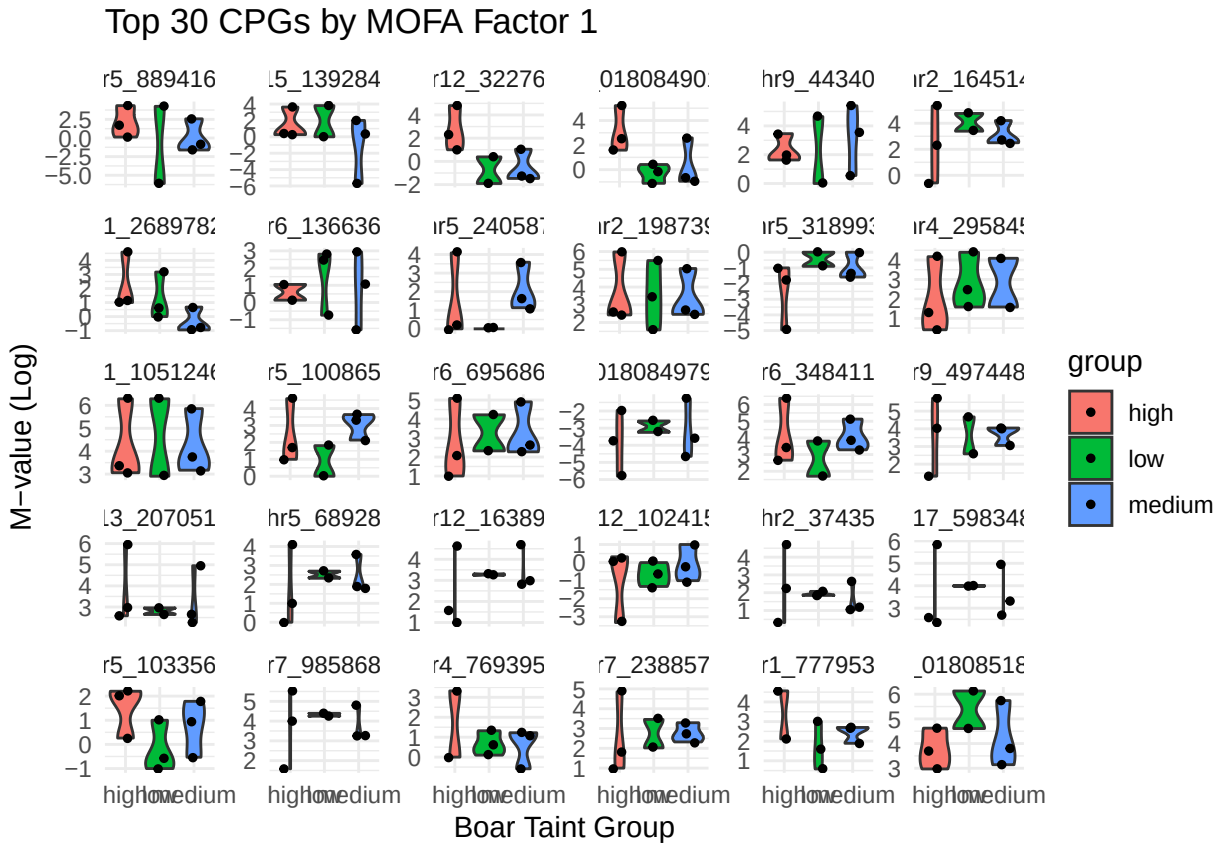
# Get input data (long format)
data <- get_data(mofa_trained, as.data.frame = TRUE)

# Subset data for top features
top_gene_data <- subset(data, feature %in% top_genes_names & view == "RNAseq")
top_cpg_data <- subset(data, feature %in% top_cpgs_names & view == "Methylation")


# Plot Genes
ggplot(top_gene_data, aes(x = group, y = value, fill = group)) +
  geom_violin(alpha = 1) +
  geom_jitter(width = 0.2, size = 1) +
  facet_wrap(~ feature, scales = "free_y") +
  theme_minimal() +
  labs(title = "Top 30 Genes by MOFA Factor 1",
       y = "Expression", x = "Boar Taint Group")
```



```
##### For CPGs
ggplot(top_cpg_data, aes(x = group, y = value, fill = group)) +
  geom_violin(alpha = 1) +
  geom_jitter(width = 0.2, size = 1) +
  facet_wrap(~ feature, scales = "free_y") +
  theme_minimal() +
  labs(title = "Top 30 CPGs by MOFA Factor 1",
       y = "M-value (Log)", x = "Boar Taint Group")
```



14. Acknowledgement

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For any further questions, feel free to contact me in the following contacts: Email: adam.elidrissi@unicatt.it - tnewnis2@gmail.com